

Laboratory & Workshop

REPIF

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Workshop

PVC holder

Introduction

A PVC holder is a protective sleeve used to hold and display cards, signs or other items. In this project a PVC holder will be customized to hold our screen and RaspberryPi.

Fabrication steps

As provided a plan we can progress step-by-step with the following topics.

- Drawing the hole points
- Drilling
- Deburring
- Filling
- Mounting screen and Raspberry Pi

Drawing the Hole Points

- **What to do:**
Use a ruler and a fine-tip permanent marker or pencil to mark where holes need to be drilled (for Raspberry Pi ports, screws, mounting points, screen cutout, etc.).
- **Tips:**
 - Measure twice, mark once.
 - Use a template or printed layout if available.
 - Mark centre points clearly with a small “+” or dot.

Drilling

- **What to do:**
Drill holes at the marked points using a drill bit of the exact required size.
- **Tips:**
 - Start with a small pilot hole for better accuracy.
 - Keep the drill perpendicular to the PVC surface.

- Secure the PVC sheet with clamps or in a vises before drilling.



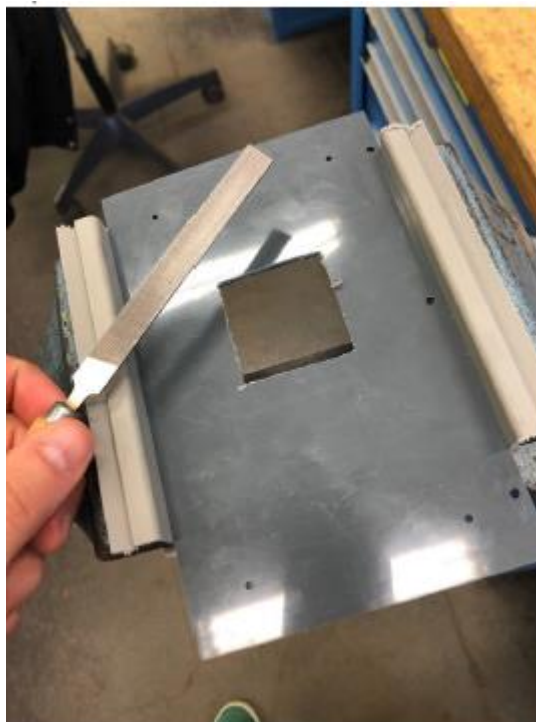
Deburring

- **What to do:**
Remove sharp edges and plastic burrs left from drilling.
- **Tips:**
 - Use a deburring tool, a small file, or fine sandpaper.
 - Smooth both the entry and exit sides of each hole.
 - Clean away plastic dust afterwards.



Filing

- **What to do:**
Smooth and shape the outer edges of the cut PVC holder.
- **Tips:**
 - Use a flat file for straight edges and a round file for curved cuts.
 - File in one direction for a cleaner finish.
 - Wipe clean to check smoothness.



Mounting screen and raspberry Pi

- **What to do:**

place the raspberry Pi on PVC holder and tighten the screws.



Mount the screen the same way on the other PVC plate as shown below and connect the cables.

Mistakes Encountered

Hole Misalignment in Early Attempts

Initially, some holes were drilled slightly off-mark (5 mm), causing misalignment with the Raspberry Pi ports.

Cause: Measuring inaccurately before drilling.



Fix: Re-measured and redrilling done.



Completion

The PVC holder for the Raspberry Pi and screen has been successfully designed, fabricated, and assembled. All planned steps—from marking and drilling holes to deburring, filing, and final mounting—have been completed according to the specifications outlined in this document.

The holder now securely houses the Raspberry Pi and display while providing necessary access to ports, adequate ventilation, and a clean, finished appearance suitable for demonstration or

daily use.



PCB

Introduction

The goal of this project was to assemble a functional PCB board by soldering electronic components onto it. The PCB acts as a mechanical support and an electrical map, using copper traces to connect parts like resistors, capacitors, and integrated circuits.

Because poor soldering can lead to "cold joints" or short circuits, care was taken to apply heat and solder precisely to each pad. This report covers the preparation of the board, the order of component placement, and the final inspection of the completed assembly to ensure it met the design specifications.

Component list

Name	Quantity
2-layer printed circuit board 75 x 85 mm	1
40-pin straight connector strip R WS40G	1
Metal resistor 0207 0.6W 10K	1
Metal resistor 0207 0.6W 56K	1
BC517 transistor	2
PCB connection terminal 2-pin RM 5	2
PCB terminal block, 2-pin RM 3.5	2
Pin strip, 1 row, straight, 1 x 3 - RM2.54 mm	1
PCB connector, straight pin header with female connector (set with cable)	1
5 mm plastic spacer roller	4
ADS1115 ADC module I2C	1
GY 302 / BH1750 Light Sensor Module	1
M2.5 Nuts	4
M2.5 Washers	4
M2.5 x 20 countersunk head	4

Fabrication Steps

As provided a plan we can progress step-by-step with the following topics.

- Checking voltage and polarity of components
- Soldering
- Attach to PVC holder
- Cables attachment

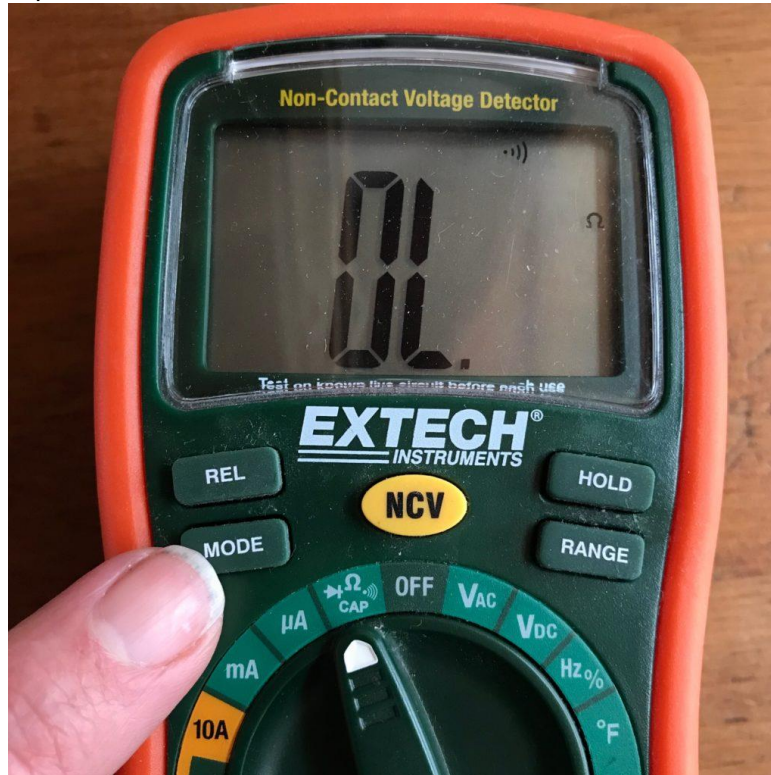
Checking voltage and polarity

- **What to do:**

Use a digital multimeter (DMM) in Continuity Mode (the icon that looks like a sound wave) to check for shorts before applying power.

 - The "Ground" Check: Place one probe on the main Ground (GND) pad and the other on the Power (VCC) pad.
 - Result: The meter should not beep. If it beeps, you have a "short circuit" (likely a solder bridge) that must be fixed before plugging it in.
 - Polarity Path: Trace from the positive input to the first component. Use the multimeter to ensure there is a continuous path where there should be, and

"open" air where there shouldn't be.



- Tips:
 - Measure twice, solder once.
 - Ask teacher to check you if possible.

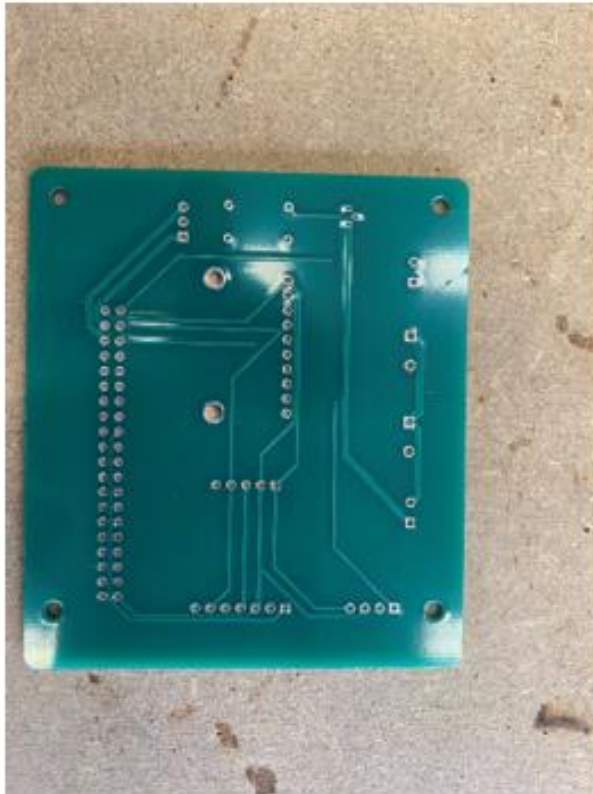
Soldering

- **What to do:**

Place the components and solder onto the board properly.

- **Tips:**

- Clean the soldering place from any dust
- Amount of melted solder must good enough
- Avoid burning PCB while soldering by keeping soldering pen too long



Attach to PVC holder

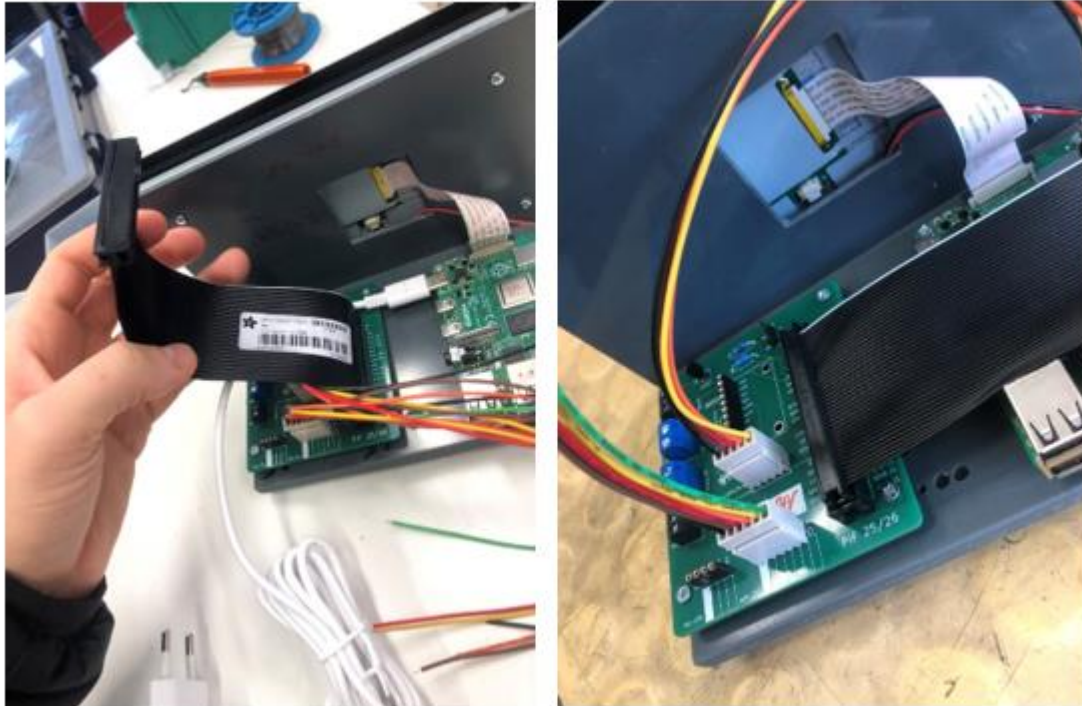
- **What to do:**
After soldering, we need to place the PCB on the PVC holder and tighten the screws.
- **Tips:**
 - You might damage the board if you tighten the screw so hard

- Some of the components have weak pins, avoid pushing hard on them when connecting the cables.



Cable attachment

- **What to do:**
According to provided sample (teacher PCB) connect given cables to the board.
- **Tips:**
 - Some of the components have weak pins, avoid pushing hard on them when connecting the cables.



Completion

The custom PCB has been successfully fabricated and populated. All components have been soldered and tested to ensure the hardware is ready for immediate operation. Designed for sensor integration and data acquisition, the board provides a reliable interface for future project phases.

With secure component placement and optimized signal paths, the board is fully operational. It includes all required peripheral and power management circuits, serving as a robust platform for

upcoming firmware development and full system integration.



Power supply

Introduction

Power supply is an essential component that provides the required energy and electricity to various devices by converting power from a source, such as an outlet or battery, into a usable form. In some cases, you may need to cut and reattach the power supply cable, whether to repair a damaged wire, extend the cable length, or replace a faulty connector. This task requires careful handling to ensure that the cable is cut cleanly and reconnected properly to maintain a safe and

efficient electrical connection. Improper handling could lead to safety risks, like electrical shorts or exposed wires, so it's important to follow the right steps and use the correct tools to ensure that the power supply is restored to full functionality.

Safety first

- Double-check connections with a multimeter to confirm there are no shorts.
- Work in a dry area to reduce the risk of electrical hazards.
- Ensure all exposed wire ends are properly insulated with tape or heat shrink tubing after soldering.

Work stops

Required tools

- Wire Cutters – For cutting the cable cleanly.
- Wire Strippers – To strip the insulation from the ends of the wires without damaging the conductor.
- Soldering Iron (if you're soldering the wires) – For securely joining wires together by melting solder.
- Soldering Wire – To use with the soldering iron to make strong, permanent connections.
- Heat Shrink Tubing and heating gun – To insulate the wires and prevent short circuits once they're reattached.

Steps

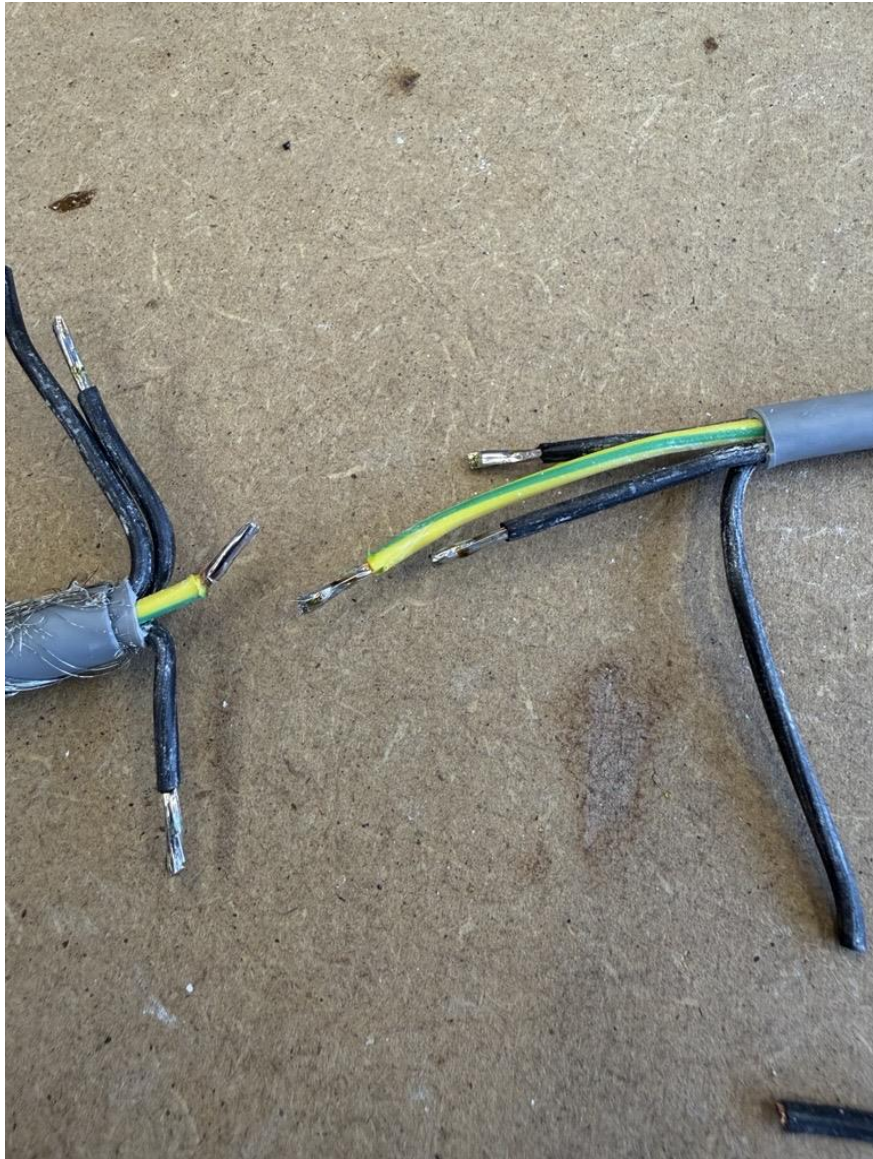
- Wire stripping: provided wire has two layer which needs to be stripped carefully. Use suitable wire stripper to avoid damage wires.



- Since the power cable contains multiple wires, they should not all be cut at the same point. Instead, each wire should be cut at a slightly different length so that, after soldering, the joints are not aligned side by side. For example, the ground wire can be soldered closer to the start of the cut, while the next wire is soldered a little further along, and so on. This staggered cutting and soldering method helps prevent the soldered joints

from touching each other, reduces the risk of short circuits, and makes the cable safer and more reliable once it is insulated.

- Apply a small amount of melted solder to the exposed wire ends before joining them, as this makes the soldering process easier and helps create a stronger, more reliable electrical connection.



- Before starting soldering we must add the tube then we can continue with soldering



- Use heating gun to give heat until tubes shrink as much as possible.



Conclusion

In conclusion, cutting and reattaching a power supply cable requires careful attention to safety and technique. By using the right tools, following proper procedures, and ensuring all connections are secure and insulated, you can restore the power supply safely and effectively. Always prioritize safety by turning off power, working in a dry environment, and checking your work to prevent electrical hazards. With the correct approach, this task can be completed with confidence, ensuring reliable functionality of the electrical system.

Sensors – A4(16bit convertor and gas module)

Introduction

The goal of this project was to assemble a functional PCB board by soldering electronic components onto it. The PCB acts as a mechanical support and an electrical map, using copper traces to connect parts like resistors, capacitors, and integrated circuits.

Because poor soldering can lead to "cold joints" or short circuits, care was taken to apply heat and solder precisely to each pad. This report covers the preparation of the board, the order of component placement, and the final inspection of the completed assembly to ensure it met the design specifications.

List of Materials

Name	Quantity
MQ135	1
ADS1115 16-Bit	1
Pin headers	Total of 10 pin heading
Soldering station and solder	

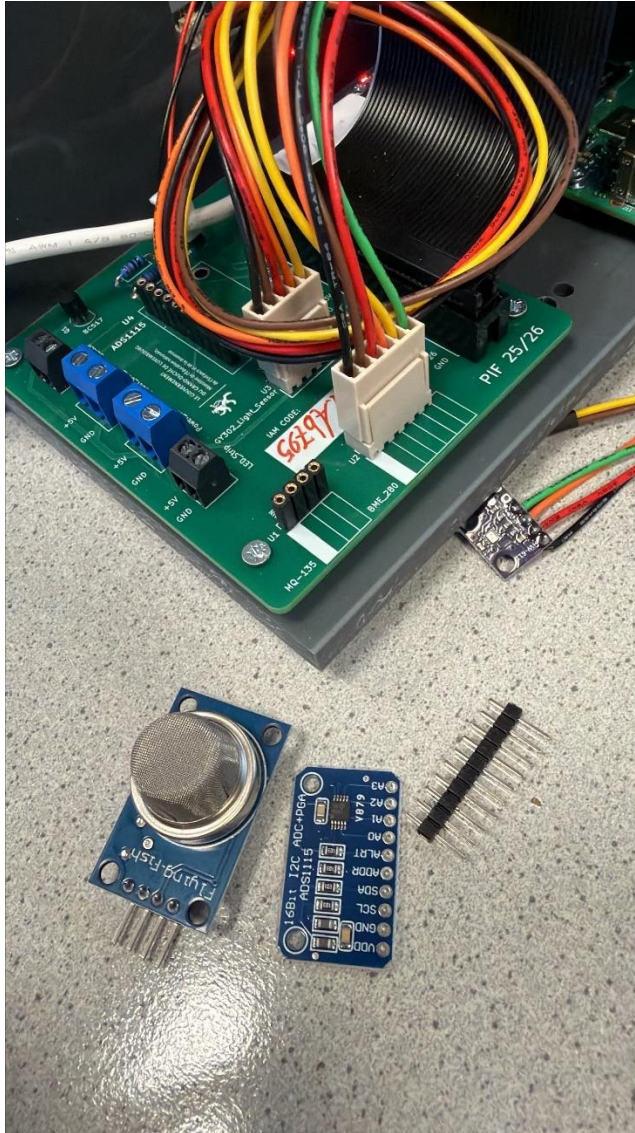
Fabrication Steps

as provided a plan we can progress step-by-step with the following topics.

- Check the necessary components
- Soldering
- Assembly

Check the necessary components

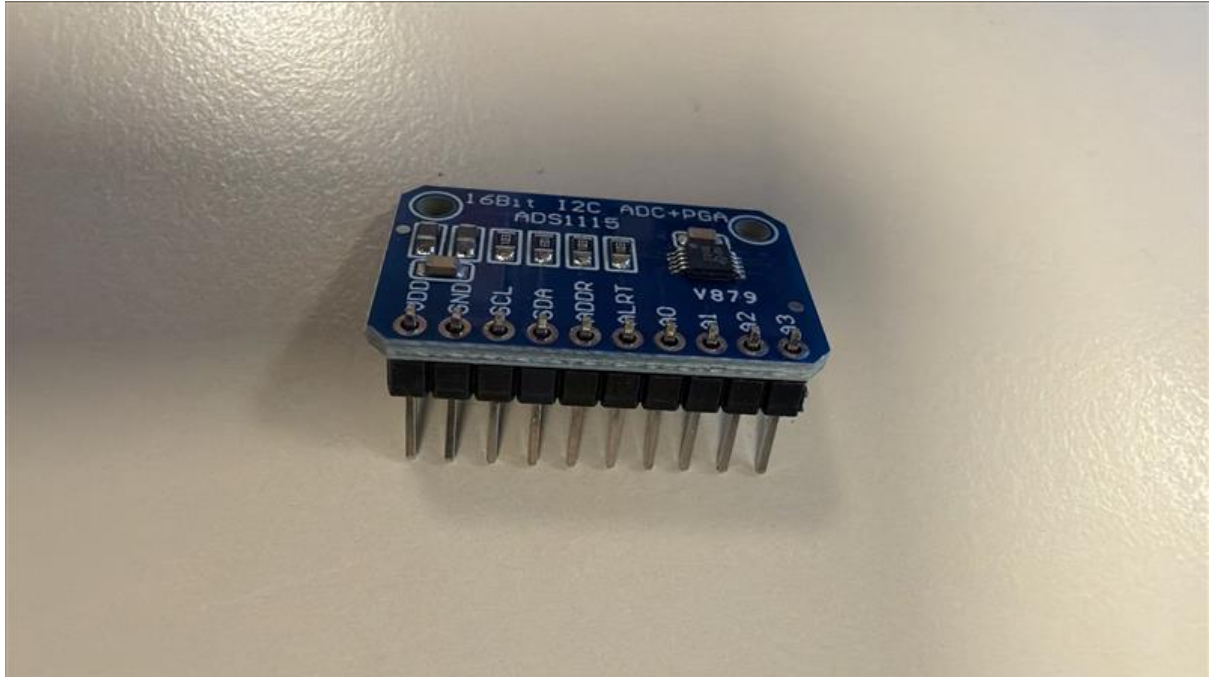
- **What to do:**
gather all the needed components and ensure noting missing.



Soldering

- **What to do:**

Solder the 10 pin headers to ADS1115 and install it properly.



Tip:

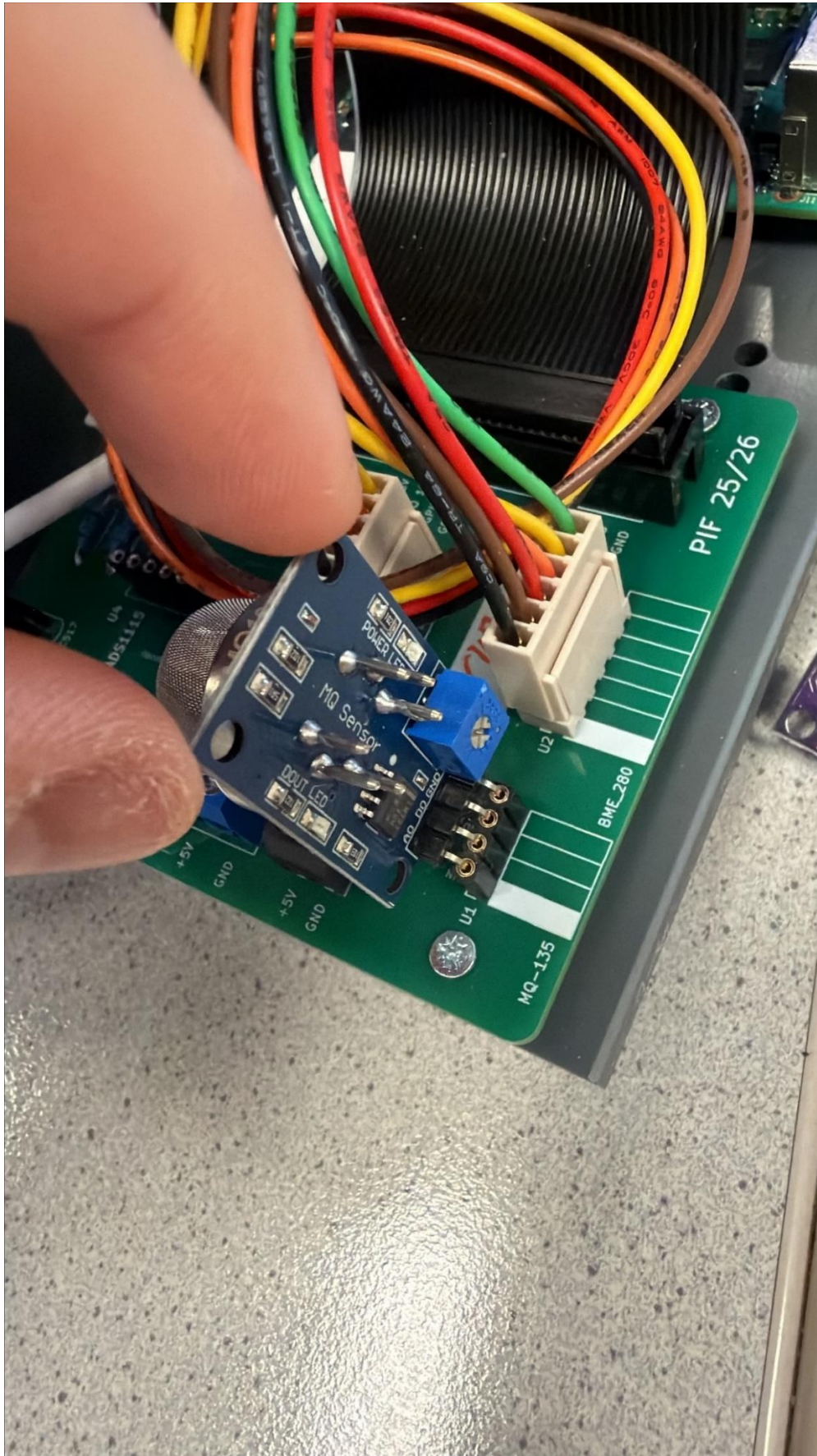
- You might burn the the sensor by keeping soldering pen for too long on the sensor
- It is recommended to use a holder to help you during soldring for right alignment.

Assembly

What to do:

Install the gas sensor gently since it has thin pins and twist its screw until green light

turns on.



Tip:

- Since pins on the PCB plate are very weak, plug the socket gently so no damage occurs.

Laboratory steps

Installation of I2C tools

Detection of I2C devices requires installing specific tools on our Raspberry pi.

Sudo apt-get install i2c-tools

```
ahmab795@AhmAb795:/opt/station $ sudo apt-get install i2c-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
i2c-tools is already the newest version (4.4-2).
The following packages were automatically installed and are no longer required:
  criu docker-buildx libcompel1 libintl-perl libintl-xs-perl
  libmodule-find-perl libnet1 libproc-processtable-perl libprotobuf-c1
  libsort-naturally-perl libterm-readkey-perl libwlroots-0.18 needrestart
  python3-protobuf python3-pycruu tini
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 18 not upgraded.
ahmab795@AhmAb795:/opt/station $
```

Detection of I2C devices

Use the following command to detect all connected i2C devices

Sudo i2cdetect -y 1

```
ahmab795@AhmAb795:/opt/station $ i2cdetect -y 1
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
20:  --  --  --  23  --  --  --  --  --  --  --  --  --  --  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
40:  --  --  --  --  --  --  --  --  48  --  --  --  --  --  --  --
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
60:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
70:  --  --  --  --  --  --  76  --  --  --  --  --  --  --  --  --
ahmab795@AhmAb795:/opt/station $
```

As you can see above there are 3 connected devices detected. After some analysis we understand the following

Device ID	Device name
0x23	BH1750(light sensor)
0x48	ADS1115(used for MQ-135)
0x76	BME280(environmental sensor)

Hopefully, this approves all our connected devices has no physical issue.

Configuration of the files

To initialize the necessary sensors data, we need to modify the (config.toml) files. This file can be found in '/opt/stations'


```

ahmab795@AhmAb795: ~
File Edit Tabs Help
GNU nano 8.4 /opt/station/config.toml
#station_serial = # Serial number of the station

interval = 2 # Time between measurements

toggle_intensity = 100 # The light level at which the LED should b>

[i2c] # I2C-Addresses of the sensors
bme280 = 0x76
bh1750 = 0x23
mq135 = 0x48

[destination] # Information about the server
ip = AhmAb795 # IP-Address or Hostname of the Ser>
path = /opt/station/measure.py # Path to the script that accepts t>

[i2c]
bme280 = 0x76
bh1750 = 0x23
mq135 = 0x48

```

The firmware will successfully initialize and read the sensors by adding the ID of devices in this file.

Testing sensors

There is already a test python file provided by teachers which allow us to test functionality of our gas sensor. Adding the sensor bus ID to the script enable us to ensure the perfection of our job. Use the following address to start file modification

```

Sudo nano /opt/stations/test-mq135.py
GNU nano 8.4 test-mq135.py
ADDRESS = 0x48 # TODO: Change this

from mq135 import MQ135
import smbus
def main():
    bus = smbus.SMBus(1) # Use 1 for Raspberry Pi, 0 for older models
    sensor = MQ135(bus, ADDRESS)

```

Run the script after the modification to see if any value obtains by sensor.

```
CO2 concentration: 3723 ppm
ahmab795@AhmAb795:/opt/station $ python test-mq135.py
CO2 concentration: 3650 ppm
ahmab795@AhmAb795:/opt/station $ python test-mq135.py
CO2 concentration: 3641 ppm
ahmab795@AhmAb795:/opt/station $ python test-mq135.py
CO2 concentration: 3629 ppm
ahmab795@AhmAb795:/opt/station $ python test-mq135.py
CO2 concentration: 3453 ppm
ahmab795@AhmAb795:/opt/station $ python test-mq135.py
$CO2 concentration: 4693 ppm
ahmab795@AhmAb795:/opt/station $ python test-mq135.py
CO2 concentration: 4563 ppm
ahmab795@AhmAb795:/opt/station $ python test-mq135.py
```

As expected, sensor returns continuously new values.

Mistakes Encountered

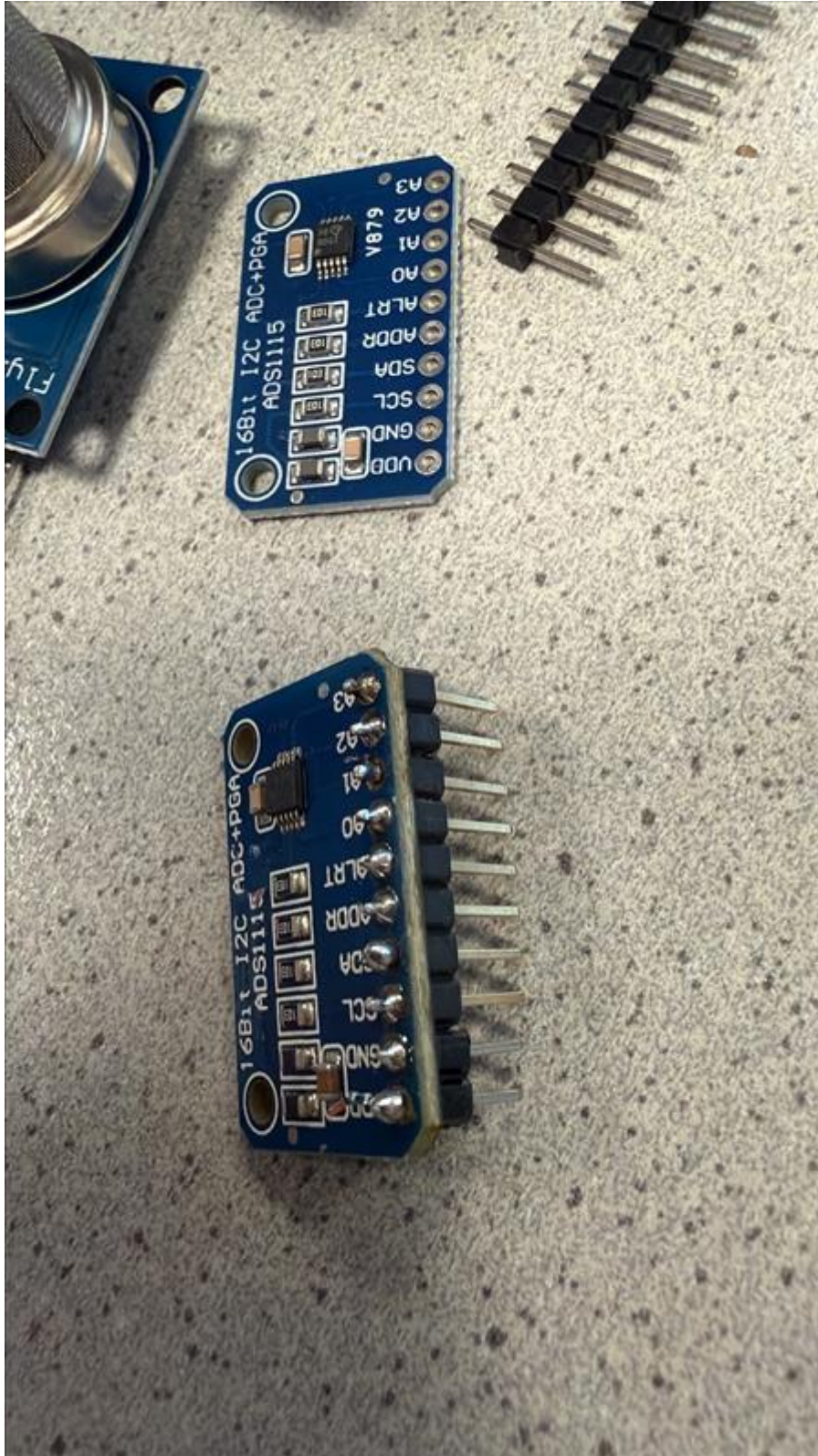
Soldering headers on the wrong side of ADC sensor

- Initially, the ADC sensor headers soldered on the wrong side.
- Fix: remove the headers and resolder.

Burning ADC sensor

- Soldering pen damaged ADC sensor when removing initial solders cause to buy new sensor and passing assignment deadline

- Fix: Purchasing new ADC sensor



Conclusion

Overall, all two sensors were installed and configured correctly and gas values were obtained successfully. Now, our system is able to keep track of gas values and hopefully we will be able to store those values and demonstrate them soon.

Sensors – A5(Barometric sensor)

Introduction

The goal of this project was to assemble a functional PCB board by soldering electronic components onto it. The PCB acts as a mechanical support and an electrical map, using copper traces to connect parts like resistors, capacitors, and integrated circuits.

Because poor soldering can lead to "cold joints" or short circuits, care was taken to apply heat and solder precisely to each pad. This report covers the preparation of the board, the order of component placement, and the final inspection of the completed assembly to ensure it met the design specifications.

List of Materials

Name	Quantity
BME280	1
BH1750	1
Socket cables	2
Soldering station and solder	
Heat Shrink Tubing	

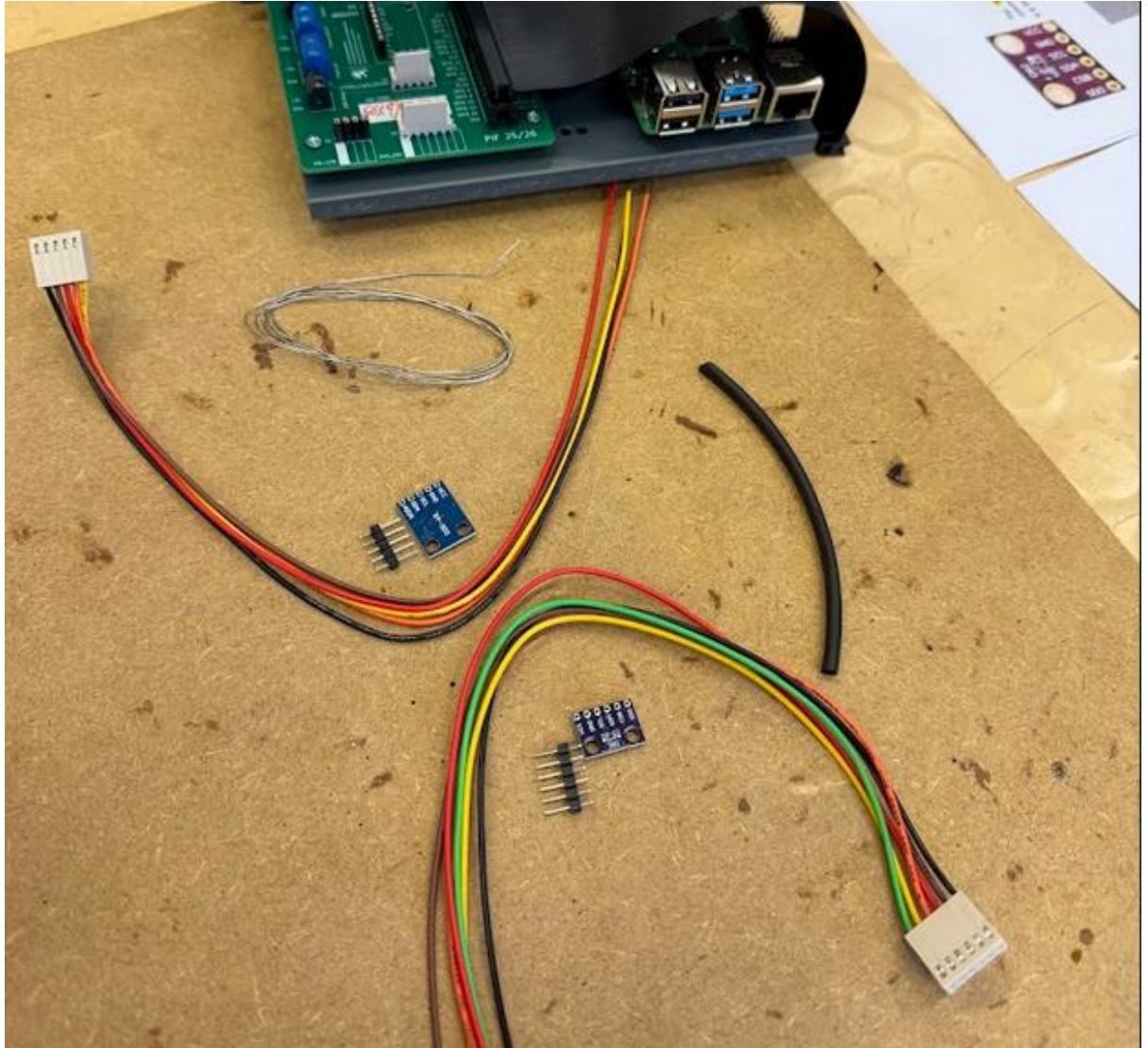
Fabrication Steps

as provided a plan we can progress step-by-step with the following topics.

- Check the necessary components
- Soldering
- Assembly

Check the necessary components

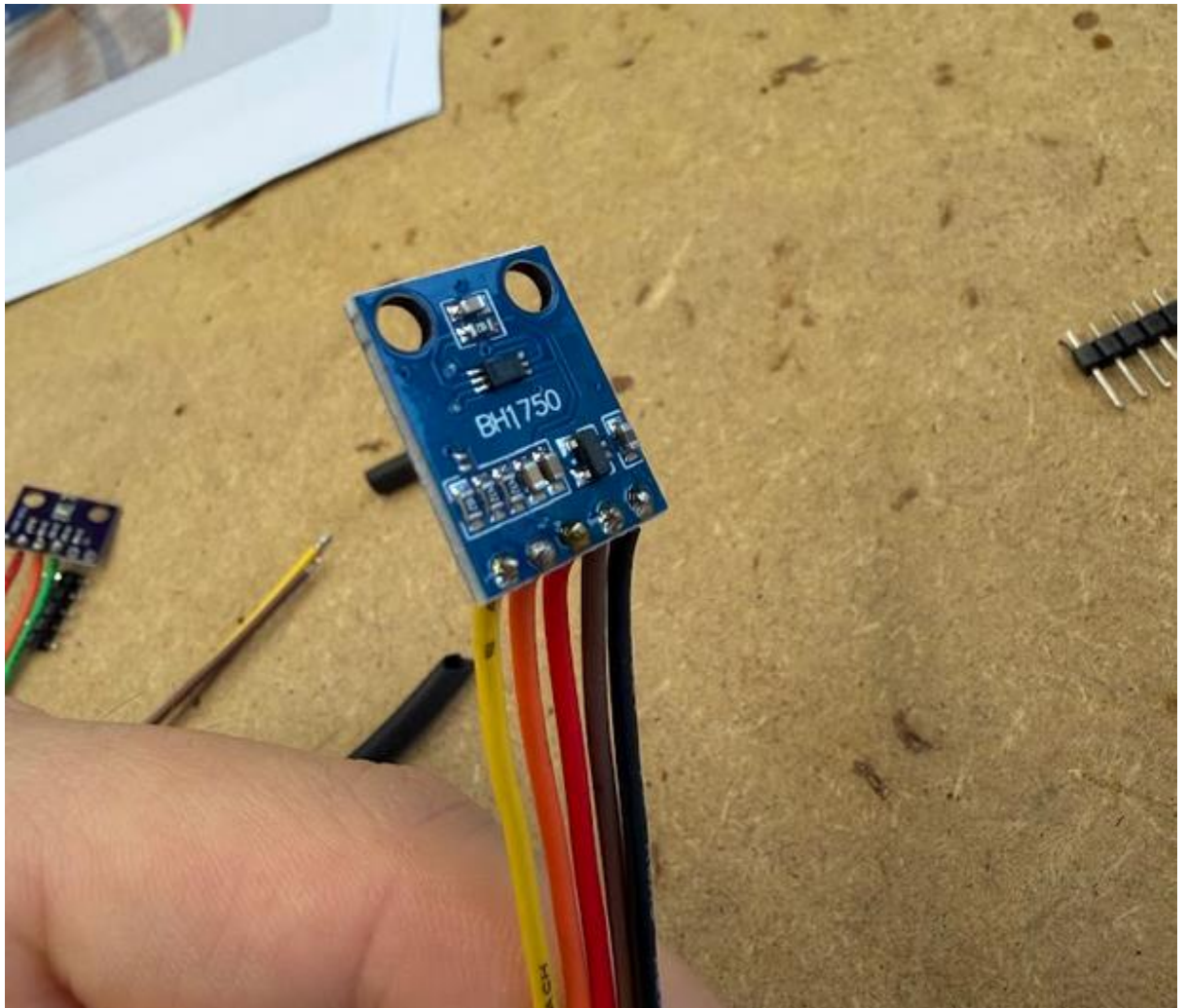
- **What to do:**
gather all the needed components and ensure noting missing.



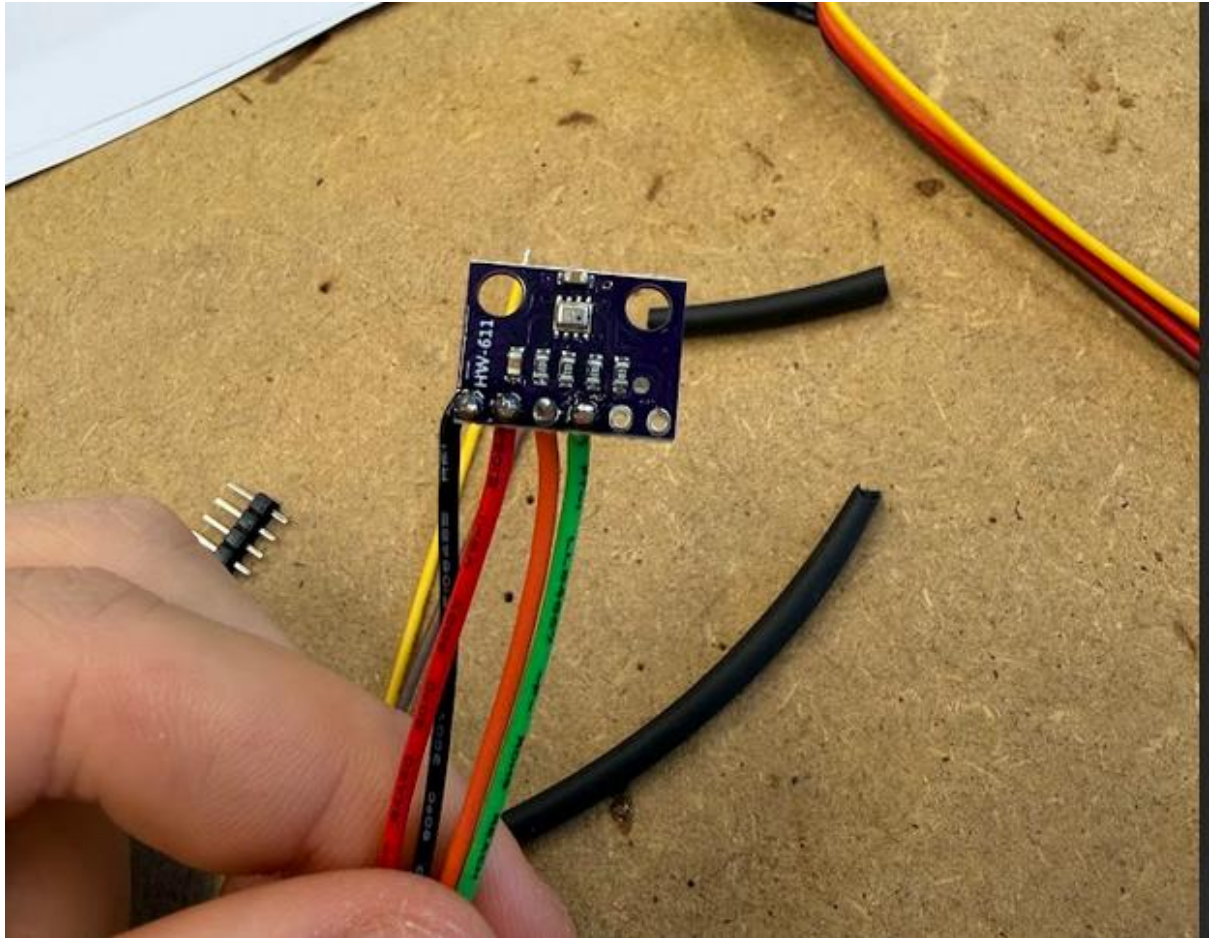
Soldering

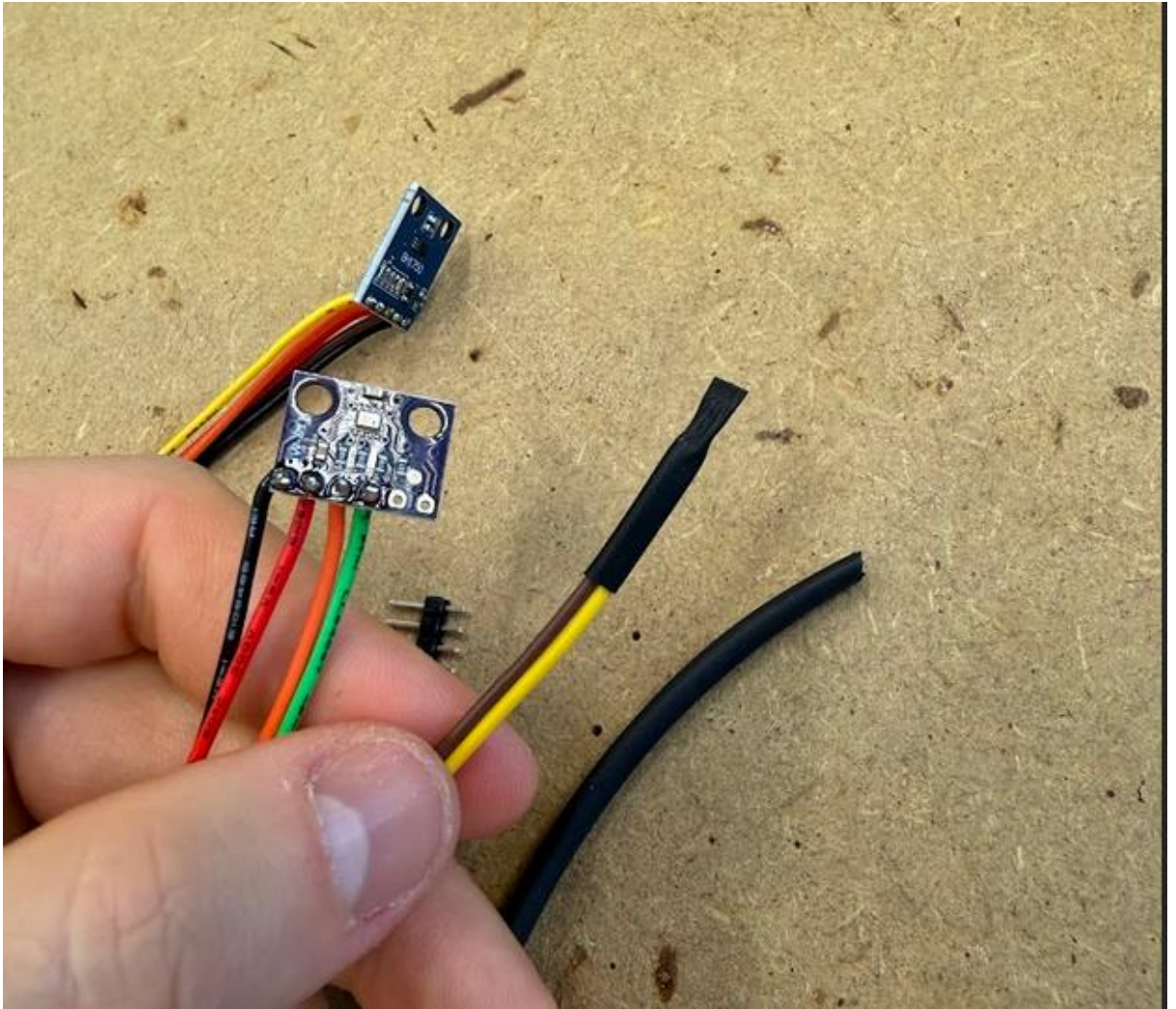
- **What to do:**

Solder the socket strip (the one that have five cable) to the sensor BH1750 as shown below.



Do the same thing for the BME280 sensor with the other cable. Keep in mind that there are two cables (yellow and brown) that needs to be covered with tube and heated with heating gun.



**Tip:**

You might burn the the sensor by keeping soldering pen for too long on the sensor

Assembly

- **What to do:**
install the soldered sockets on the PCB board.



Tip:

Since pins on the PCB plate are very weak, plug the socket gently so no damage occurs.

Mistakes Encountered

Fail to receive values from sensors

BMP280 Sensor

- Despite images show the quality of soldering but still receiving measures encounter problem.
- Fix: we must replace BME280 with BMP280 python library. However, since Debian-based system do not allow pip to modify the system python we need to install it by breaking the

system using the following command:

```
ahmab795@AhmAb795:/opt/station $ python3 -m pip install bme280 --break-system-packages
```

BH1750 Sensor

- I recheck the soldering and ensure that the problem is the sensor itself.

```
ahmab795@AhmAb795:/opt/station $ python3 test-bh1750.py
Light Level: 0.0 lux
```

- I had to replace the sensor with a new one.

```
ahmab795@AhmAb795:/opt/station $ python3 test-bh1750.py
Light Level: 75.0 lux
ahmab795@AhmAb795:/opt/station $ python3 test-bh1750.py
Light Level: 74.16666666666667 lux
ahmab795@AhmAb795:/opt/station $ python3 test-bh1750.py
Light Level: 0.8333333333333334 lux
ahmab795@AhmAb795:/opt/station $ python3 test-bh1750.py
Light Level: 2.5 lux
```

Completion

The PCB holder for the Raspberry Pi and screen has been successfully designed, fabricated, and assembled. All planned steps—from marking and drilling holes to deburring, filing, and final mounting—have been completed according to the specifications outlined in this document.

The holder now securely houses the Raspberry Pi and display while providing necessary access to ports, adequate ventilation, and a clean, finished appearance suitable for demonstration or daily use.

General Assembly

Introduction

The goal of this project was to assemble our station wooden box. This wooden box acts as a protector for our display and raspberry pi. We planned to assemble the box and make some holes to mount sensors and light switch.

List of materials

Name	Quantity	Description
Wooden box pieces		
Switch	1	On and off switch for lighting
Wire	2	Preferably 5 cm length
Screw and nuts	4	To mount sensors
Screwdriver		
Soldering station	1	
solder		As much as needed

Fabrication Steps

as provided a plan we can progress step-by-step with the following topics.

- Marking and Making holes before mounting

- Assemble box body
- Mounting and Wiring switch



Making holes

- **What to do:**

Use a marker or pencil to mark a suitable point for your sensors and start drilling. Then, use a soft file to make round's holes less sharp(Figure 2).

- Use drill number 3 for sensors hole

Tips:

- Ask a friend to hold you a vacuum cleaner to avoid mess

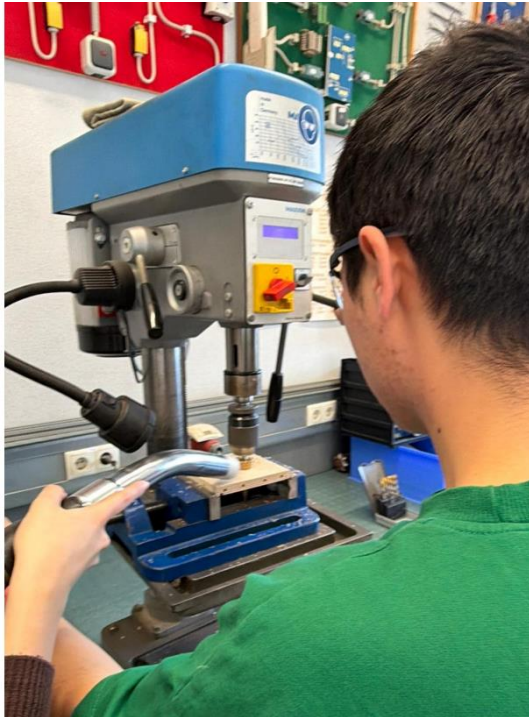


Figure 1



Figure 2

Assembling

- **What to do:**
Use plastic attacher shown below (Figure 3) to assemble box body. After assembling two sides, you can place your station inside the box.
- **Tips:**
 - Your wooden box has specific box space for screen (Figure 5)
 - Station is not supposed to touch the bottom since there is four plastic underneath

*Figure 3**Figure 4*



Figure 5

Switch connection to PVC

- **What to do:**
In this step we must solder two cables covered by tube and connect it to our PVC board.

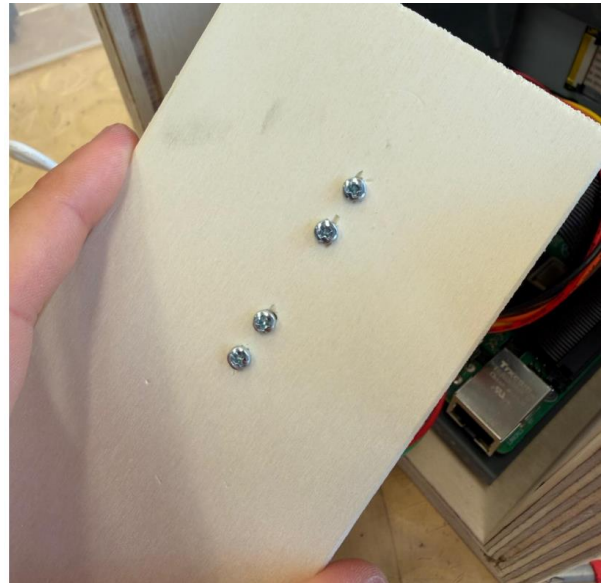
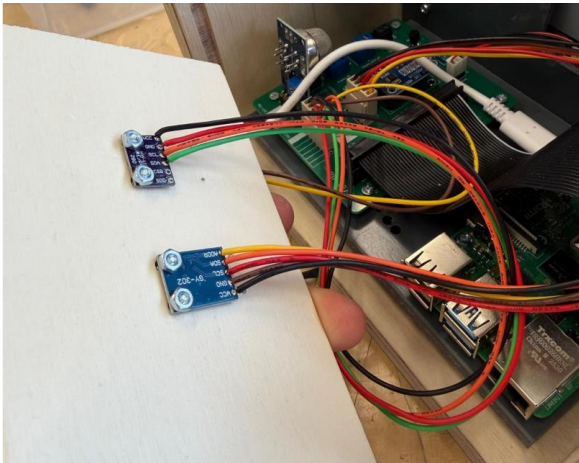


- **Tips:**
 - Place your station inside the box before installation of switch other wise switch will block station way to set properly inside the box.

Installation of sensors

What to do:

Place two sensors on the made holes and screw them



Final look



Mistakes Encountered

Broken edge of on the body pieces

- An edge of one of box body part broke accidentally and I had to replace it.
- Cause: Assembly harshly.

Completion

This body box for our Raspberry Pi plays a crucial role in our project. We made the box assemble, install switch, sensors, and station inside the box successfully. All our plans and point meet with success according to expectation.

Lighting

Introduction

The goal of this project is to connect a LED strip to our station. This LED strip should be installed in front of our station and needs to be programmed to be on and off using a switch or button.

List of materials

Name	Quantity	Description
LED strip	15 cm	Figure1.1
Drill	1	To make hole for wiring
Wire	2	Preferably 10 cm length
Glue gun	1	To attach LED strip to station
Screwdriver		
Soldering station	1	
solder		As much as needed

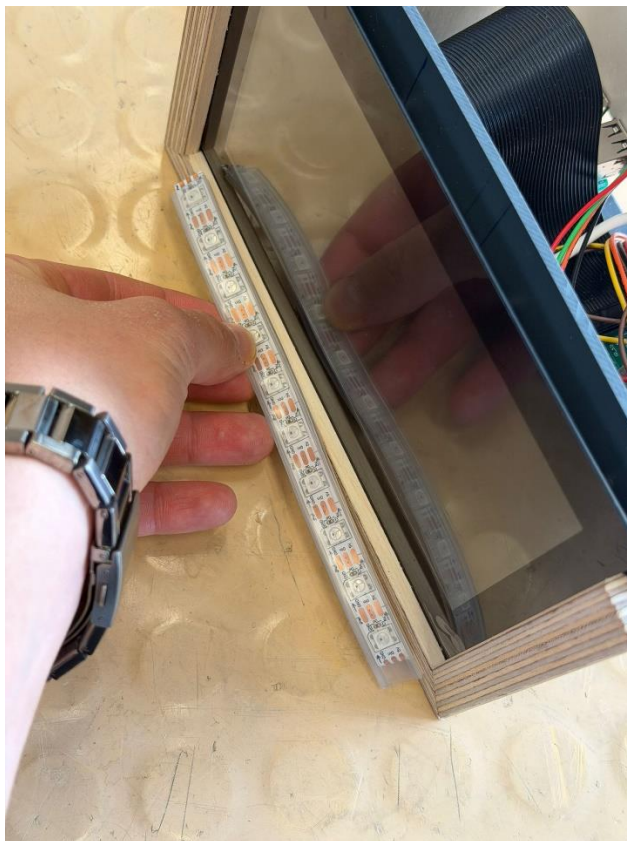


Figure 6.1

Fabrication Steps

as provided a plan we can progress step-by-step with the following topics.

- Marking and Making hole for wiring
- Solder the wire to LED strip
- Attach LED strip to your station
- Connect the LED wires to PCB

Making hole

- **What to do:**

Use a marker or pencil to mark a suitable point for your wiring and start drilling. Then, use a soft file to make round's hole less sharp(Figure 1).

- Use drill number 6 for hole

Tips:

- Ask a friend to help you while drilling to hold the plate

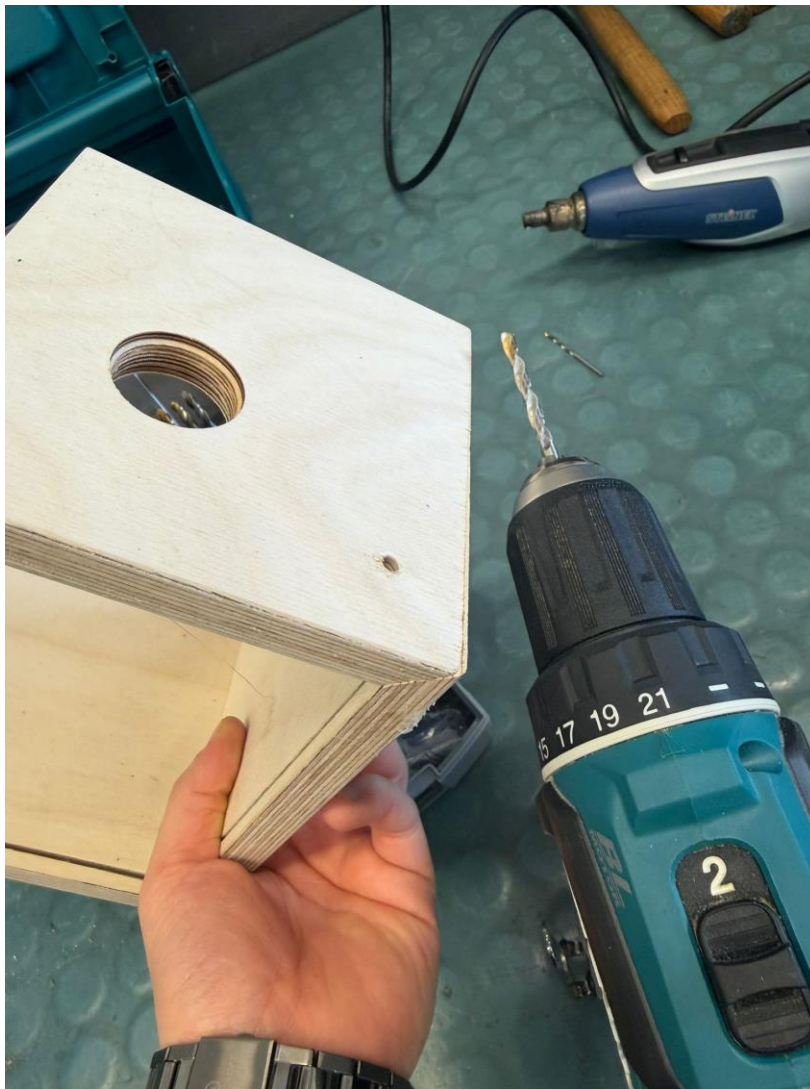


Figure 7

Soldering

- **What to do:**
Pull the LED strip out little bit from its plastic cover and solder wires to it gently (Figure2).
- **Tips:**
 - The LED strip is very sensitive about the heat so avoid long contact with soldering pen.

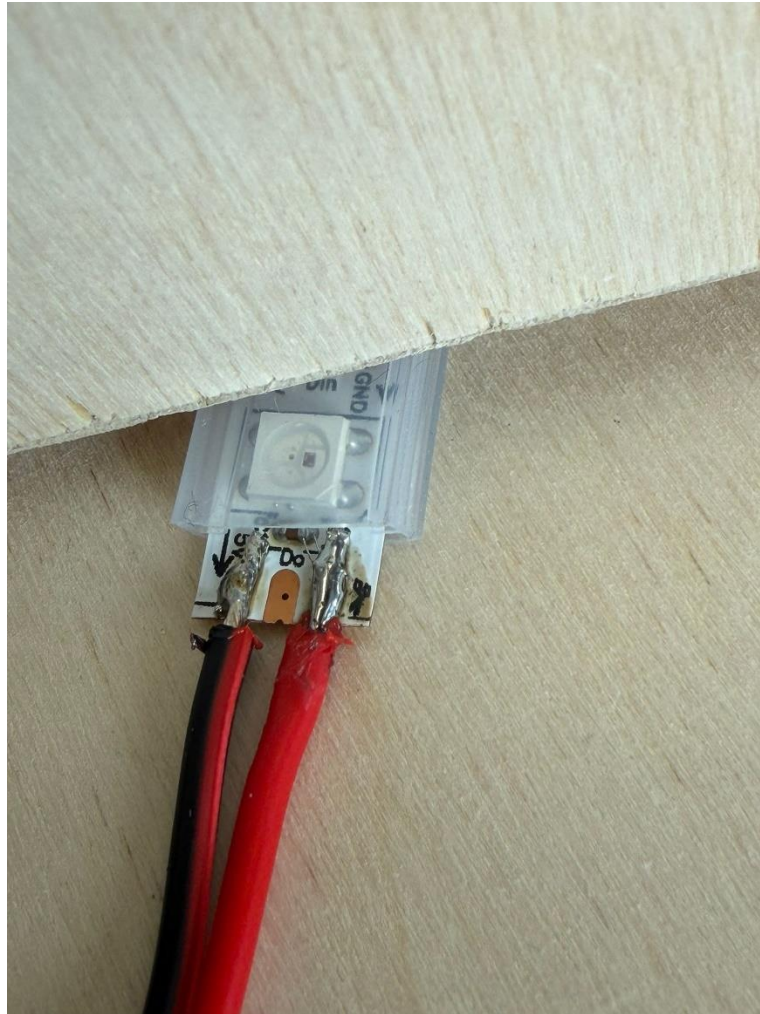


Figure 8

LED strip attachment

- **What to do:**
Connect the glue gun and wait until it reach hot enough. After reaching to proper degree you can glue target spot where you want the LED stays(Figure3).
- **Tips:**
 - Hold the LED strip on the surface with hands after applying for few second for better adhesion.



Figure 9

Connection to PCB

What to do:

pass the wires you solder to LED through the hole you made earlier and connect them to your PCB(Figure4).

Tips:

- It is better to take the device out of box and remove the button to connect the wires to PCB(Figure5,6)

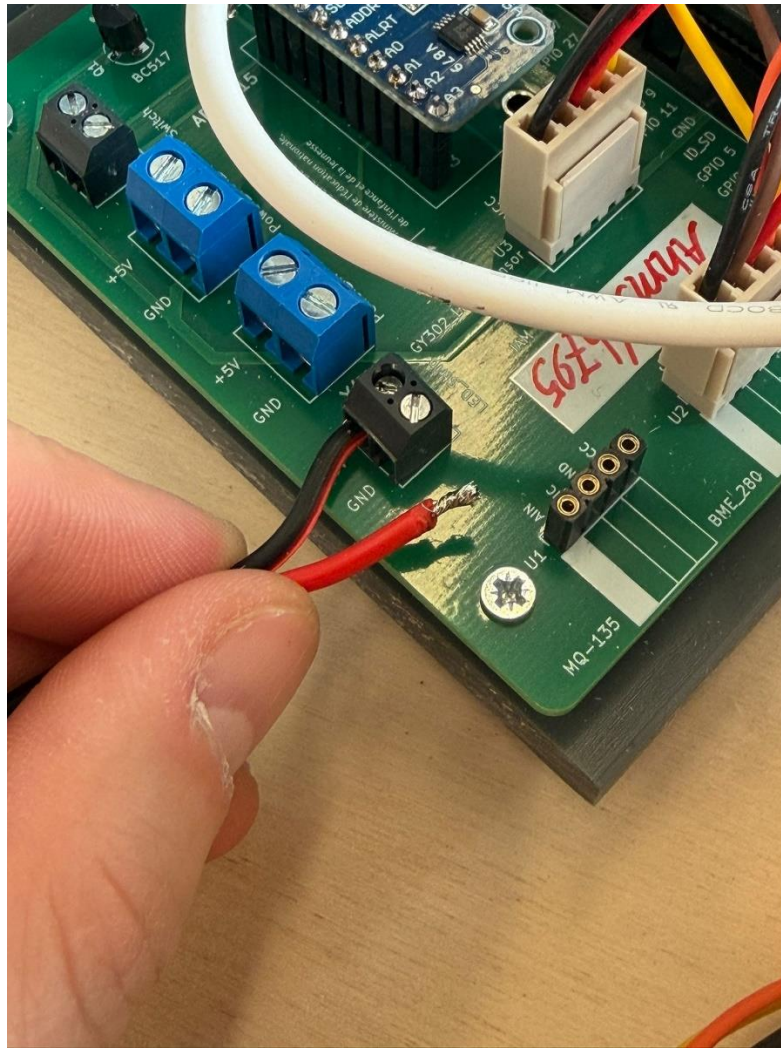
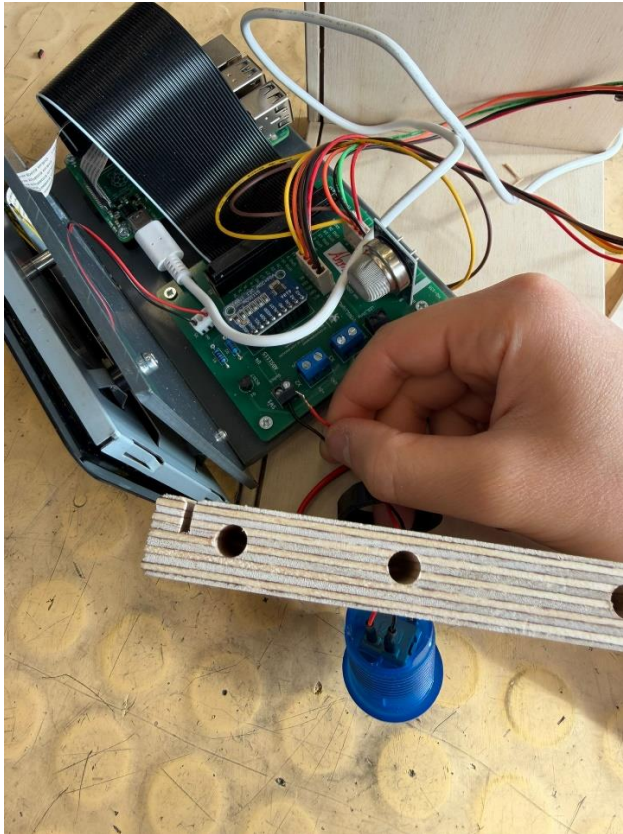
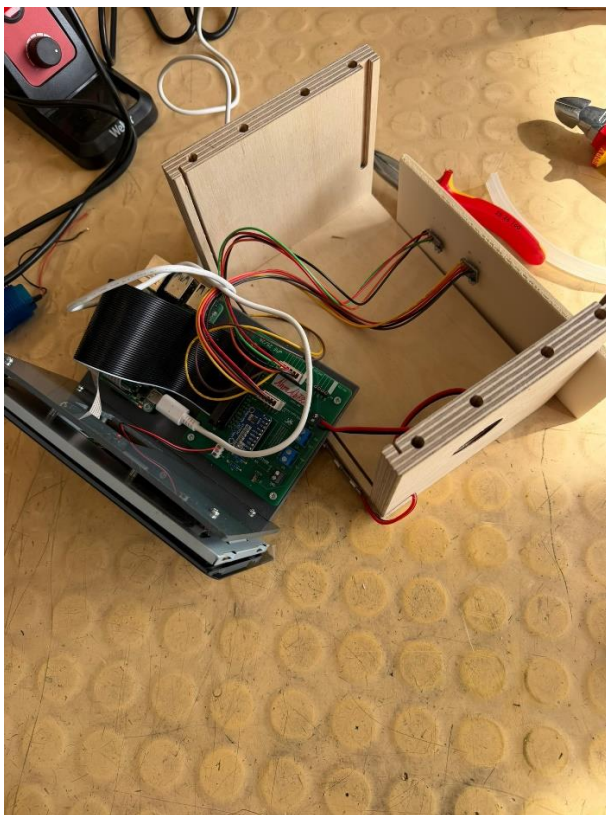


Figure 10

*Figure 11**Figure 12*

Mistakes Encountered

- I encountered a small problem with attachment of LED strip to wooden body of station. The adhesion couldn't be made since the LED cover were made of silicone and silicone were not attachable to wood using glue gun, so we had to remove the LED cover completely.

Final look



Laboratory

Raspberry Pi

Installation Raspberry Pi and enabling SSH

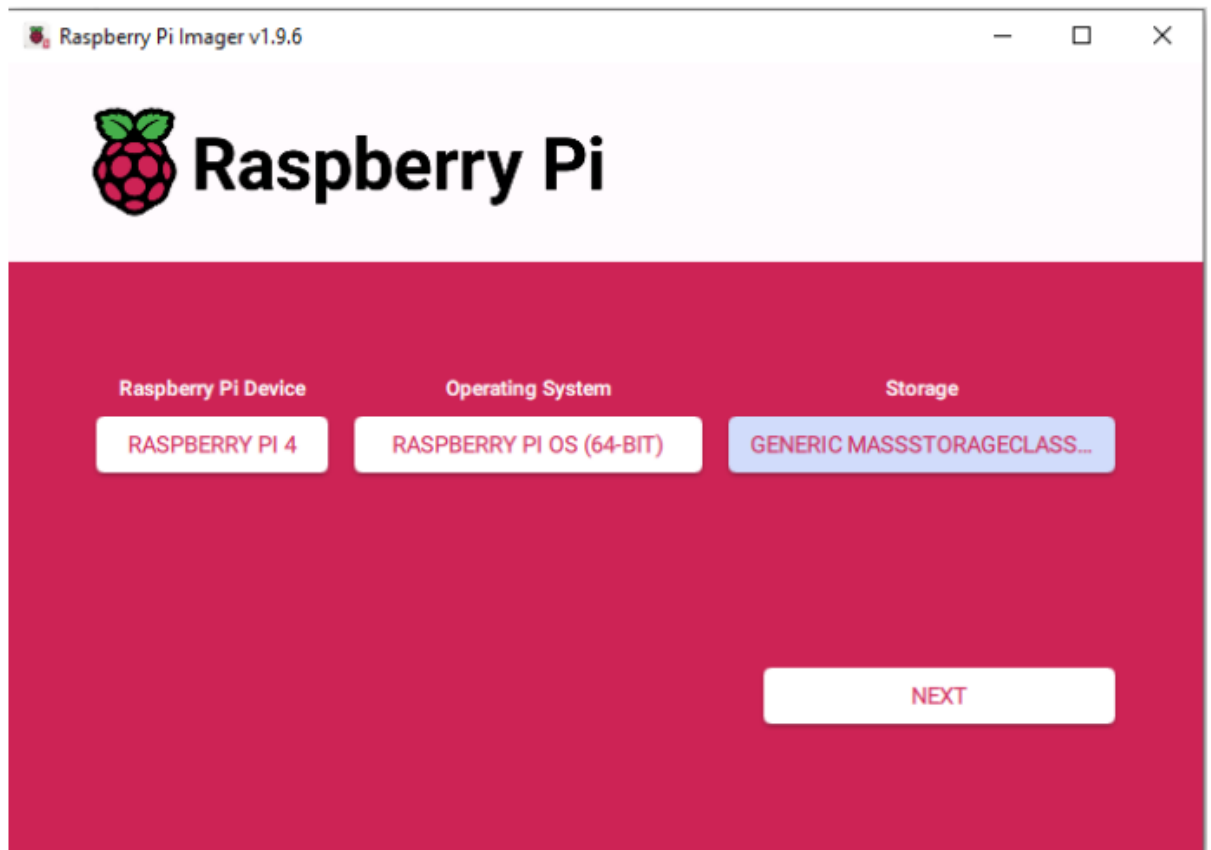
Follow these steps to install the operating system on your SD card and enable SSH for remote access.

1. Download and install Raspberry Pi Imager from the official website:
<https://www.raspberrypi.com/software/>
2. Insert an SD card into your computer and open the Raspberry Pi imager application.
3. Device, OS and Storage
Choose Device: Click "Choose Device" and select your specific model of Raspberry Pi.
Choose OS: then, click "Choose OS" and select Raspberry Pi OS (64-BIT)
Storage:
 - On the General tab
 - set your hostname for your Pi
 - Set a username and password for the default user

Configure your wireless LAN

On the Service tab, enable SSH and mark 'Use password authentication'

Option tab doesn't need any modification for our task



OS Customisation

General Services Options

☒ Set hostname: AhmAb795 .local

☒ Set username and password

Username: ahmab795

Password:

☒ Configure wireless LAN

SSID: LPEM_B13

Password:

☐ Hidden SSID

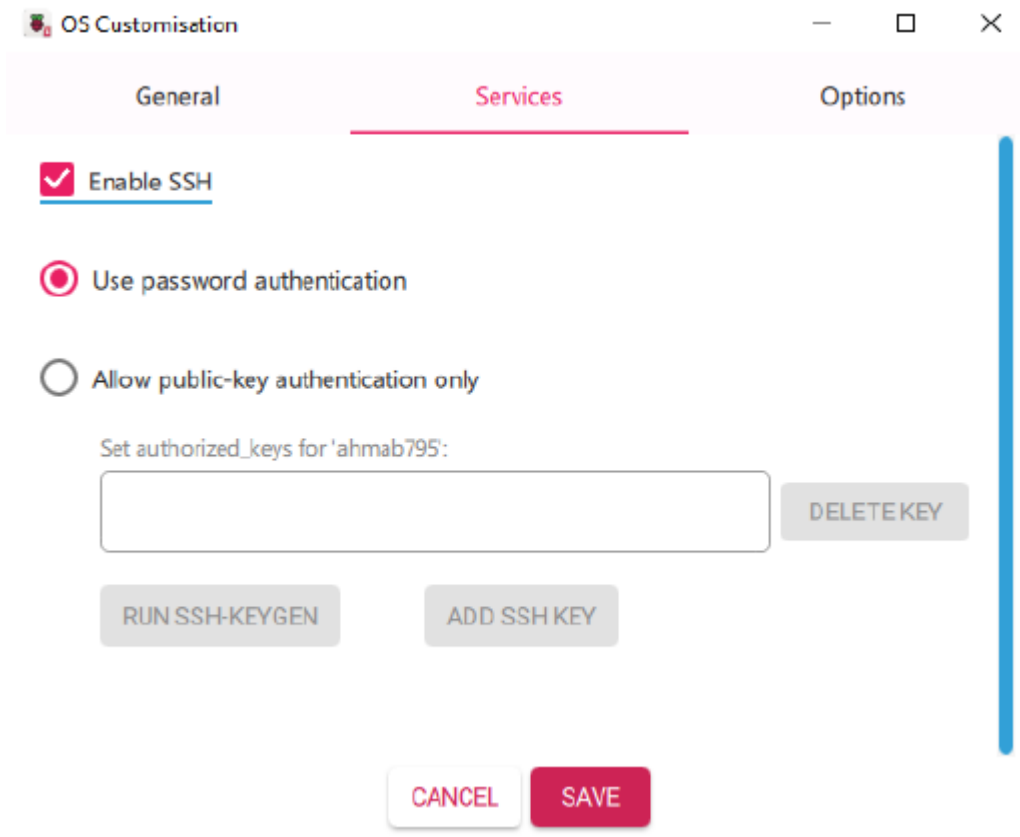
Wireless LAN country: LU

☐ Set locale settings

Time zone: Europe/Luxembourg

Keyboard layout:

CANCEL SAVE



OS Customisation

General **Services** Options

☒ Enable SSH

☒ Use password authentication

☐ Allow public-key authentication only

Set authorized_keys for 'ahmab795':

DELETE KEY

RUN SSH-KEYGEN ADD SSH KEY

CANCEL SAVE

4. Click "Save" to close and apply these settings.
5. Finally, click "Write" to begin writing the OS to the SD card. This will erase all data on the card.

Connection to VNC with premade account

Since editing `/etc/dhcpd.conf` didn't work to set a static IP, we need to install VNC on our Raspberry Pi and set the static IP manually using the graphical interface.

Here are the commands to install VNC

1. connect to Pi via ssh with pre configured hostname(AhmAb795)
2. enter the following commands

Sudo raspi-config

Then:

- Go to Interface Options → VNC → Yes, then Finish and reboot if asked.

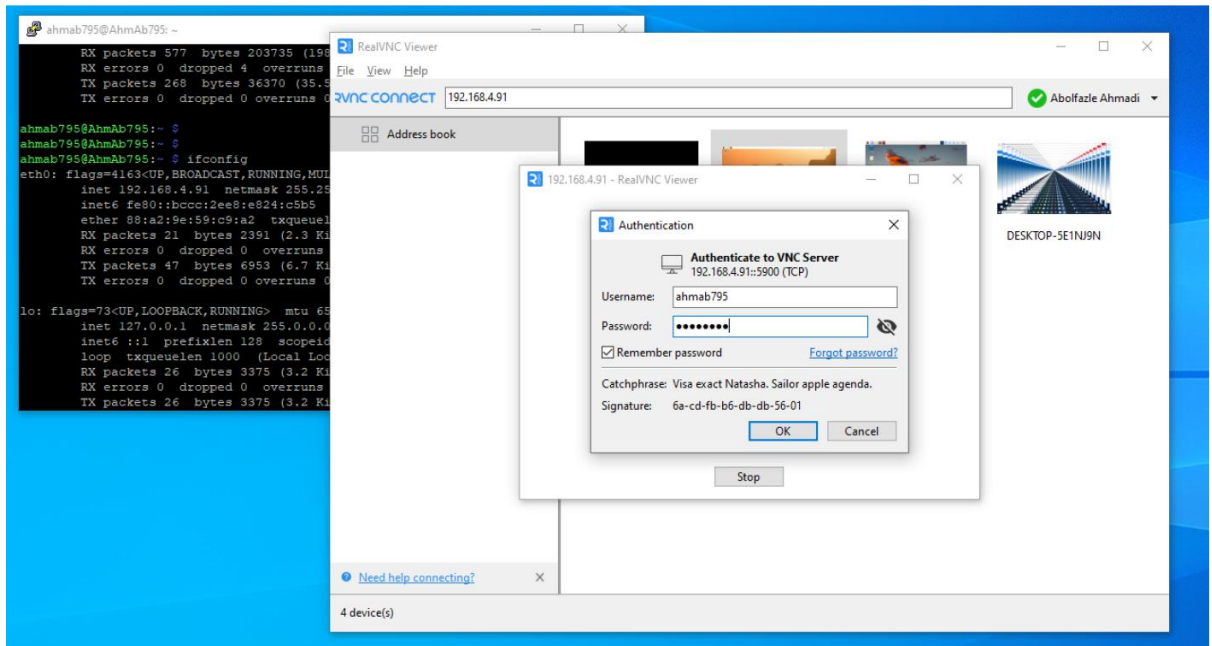
3. Create a VNC account in the official website of RealVNC

<https://www.realvnc.com/en/>

4. install the VNC by entering the following commands

```
Sudo apt update
Sudo apt install realvnc-vnc-server realvnc-vnc-viewer
Sudo systemctl enable vncserver-x11-serviced.service
```

- open VNC in you local computer and enter IP of your Pi. Then you will have access to you Pi by entering right credentials.



Configuring a Static IP Address via the Desktop (GUI)

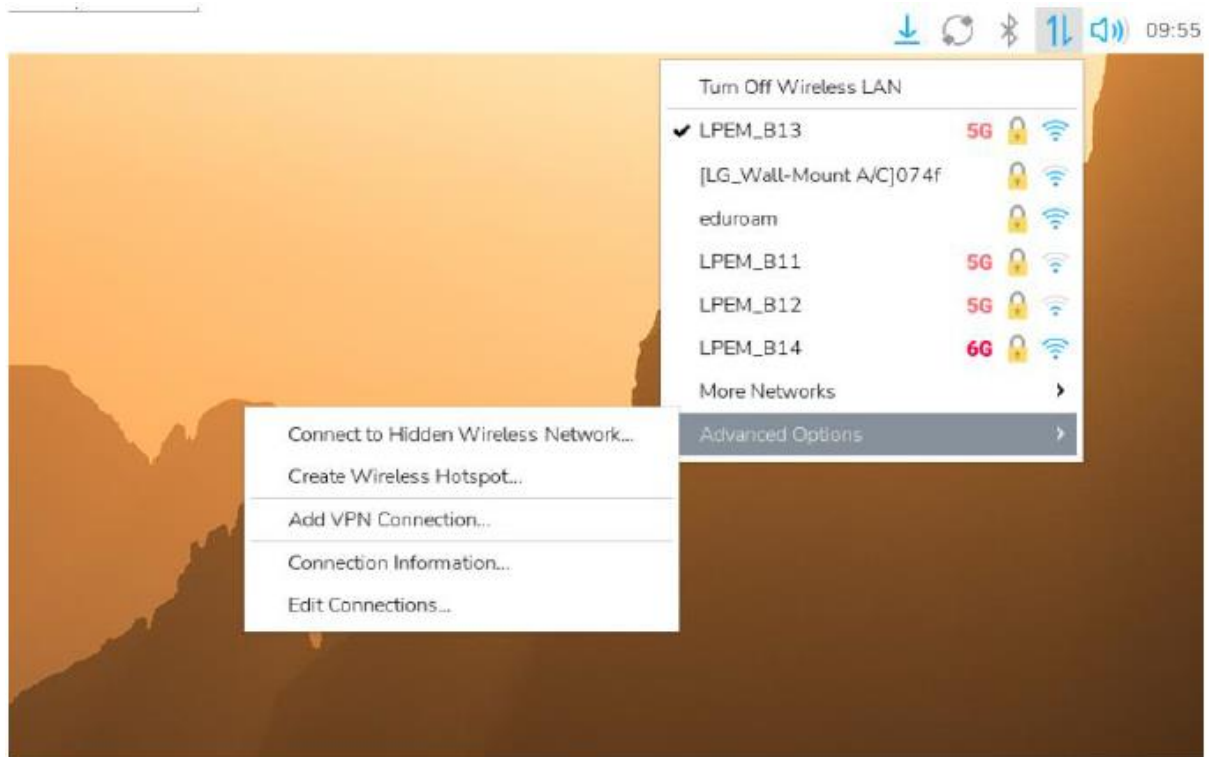
You can configure a static IP address for your Raspberry Pi based on your connection type (Wi-Fi or Ethernet).

Benefits of a Static IP Address

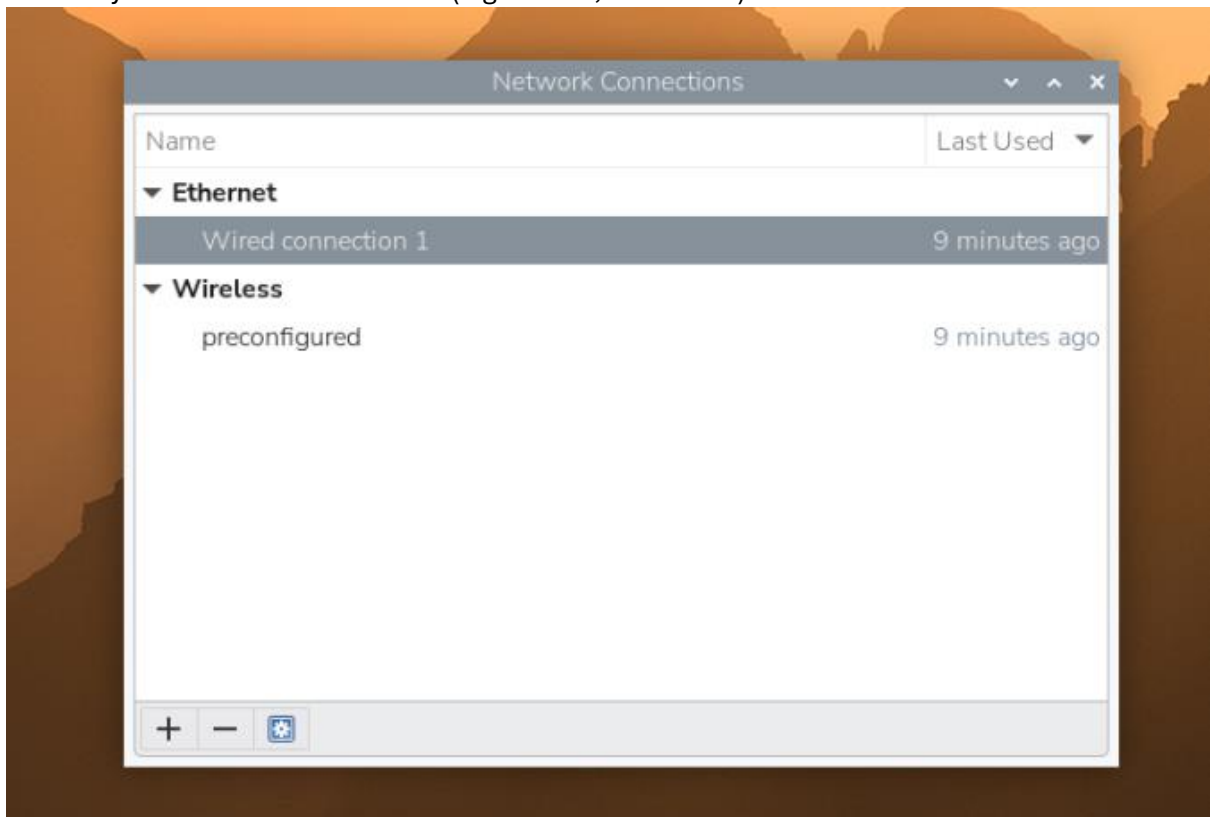
- Convenient Connections: Provides a reliable, unchanging address for SSH, VNC, and other remote access services.
- Easy to Remember: You only need to remember one IP address for your device instead of tracking changes made by the router.

Configuration of Static IP address

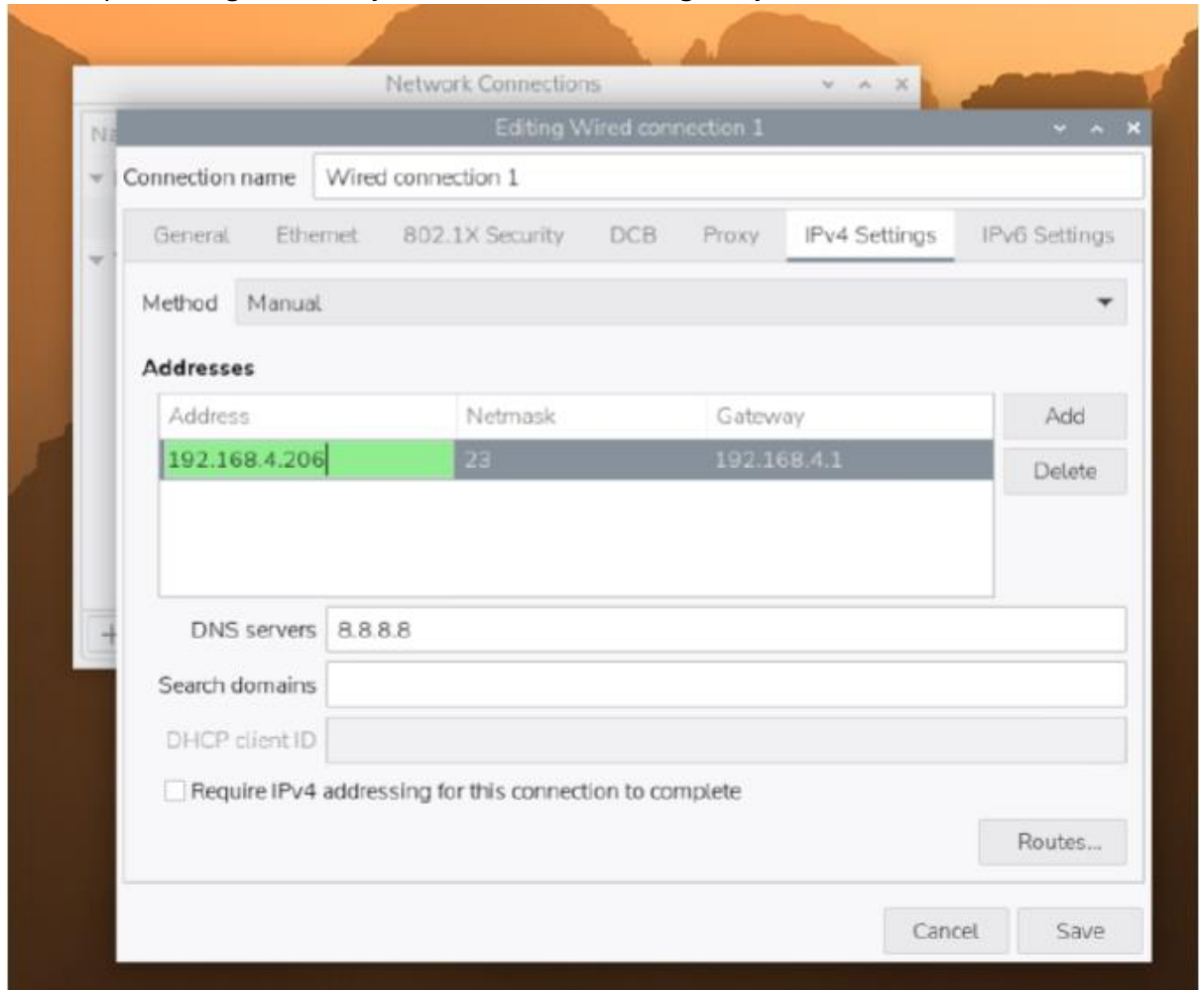
- On top right of your screen, click on double arrow for the network menu. Then navigate to “Advance Options”-“Edit connections...”



- Choose your method of connection(e.g “wired”, “wireless”)



- On the Ipv4 setting tab, enter your desired IP, netmask, gateway and DNS address

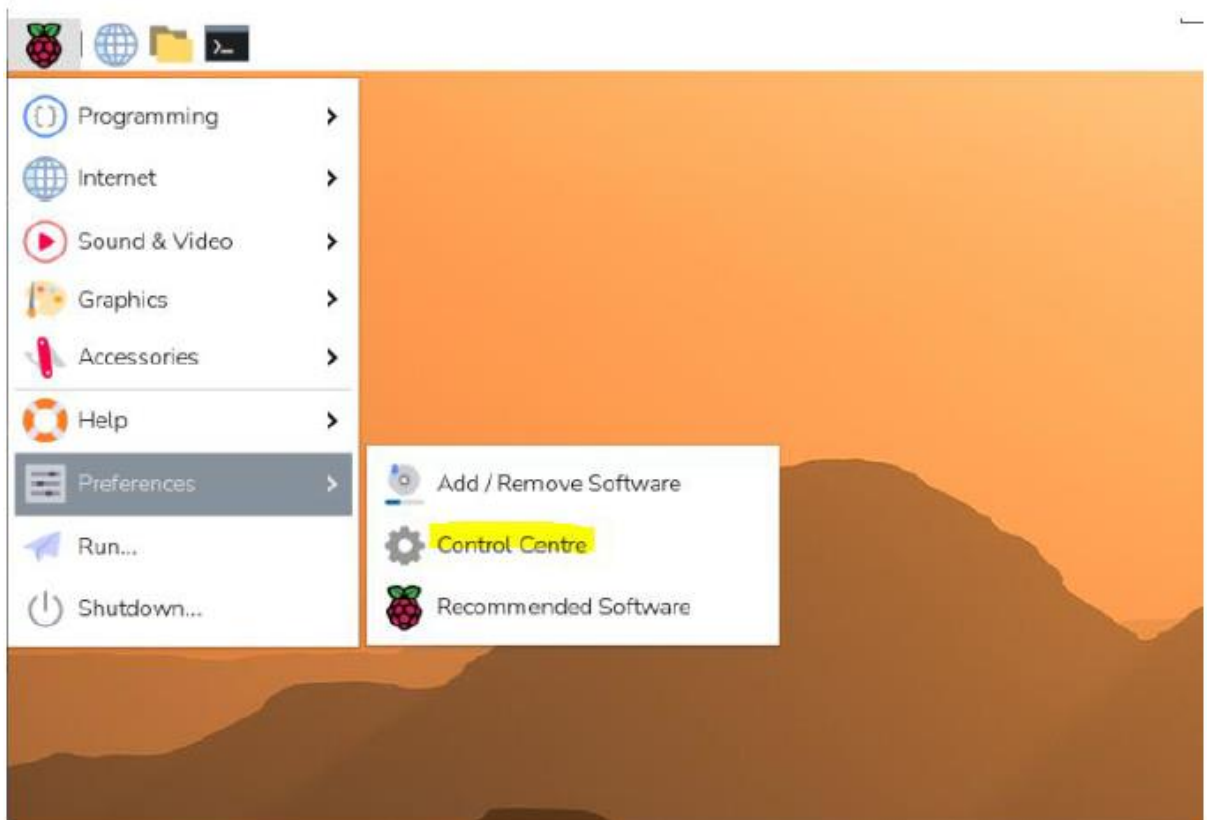


- Click Save and then close the window. You may need to disconnect and reconnect to the network for the changes to take effect. The last octet(206), could be changed as long as it doesn't go beyond the range or already taken.

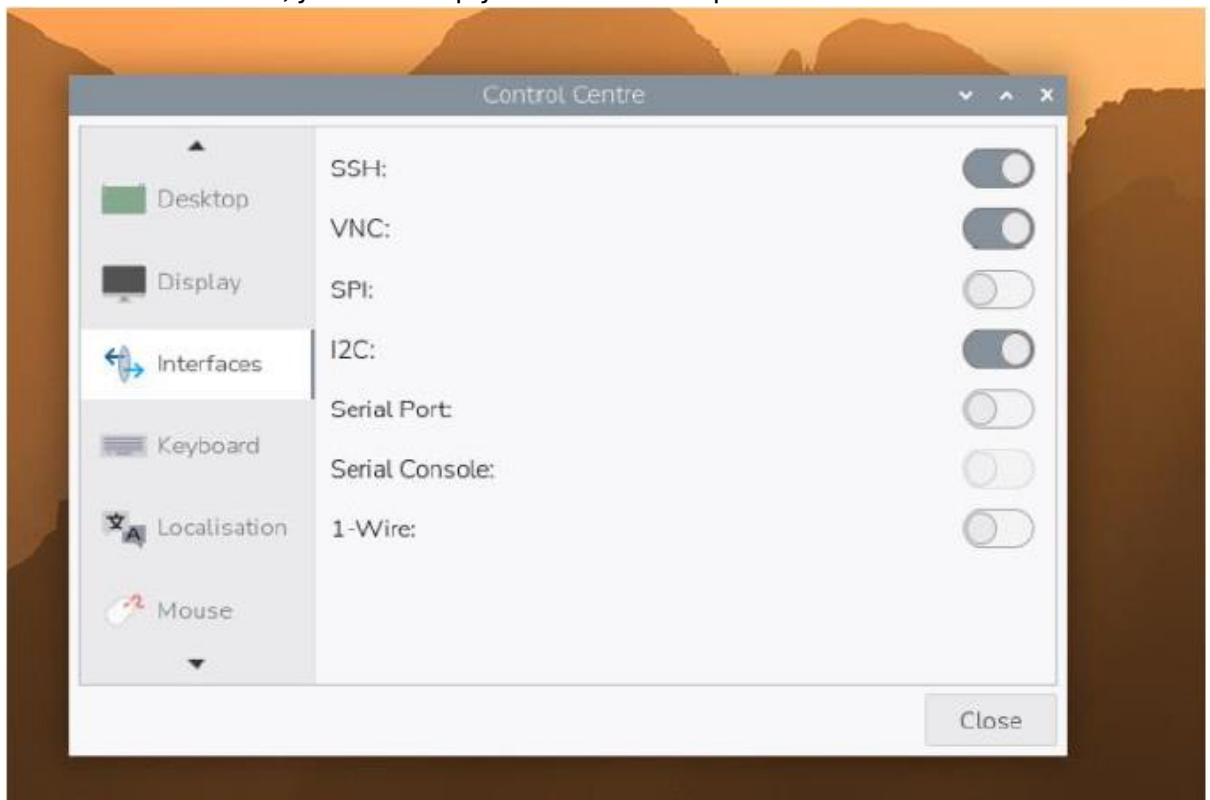
Activation of I²C interface

Follow these steps to enable the I²C interface using the Raspberry Pi configuration tool.

- On the top left of your screen, click on raspberry icon and select “Preferences” then “Control Center”.



- In the “Interface” tab, you can simply turn on the I²C option.



Installation of Docker

In short, Docker packages an application and all its dependencies into a standardized, isolated unit called a container. This solves the universal problem of "it works on my machine."

To install the latest stable versions of Docker CLI, Docker Engine, and their dependencies:

- Download the script

```
curl -fsSL https://get.docker.com -o install-docker.sh
```

- verify the script's content

```
cat install-docker.sh
```

- run the script with --dry-run to verify the steps it executes

```
sh install-docker.sh --dry-run
```

- run the script either as root, or using sudo to perform the installation

```
sudo sh install-docker.sh
```

```
Sudo systemctl status docker
```

- Check status of Docker service

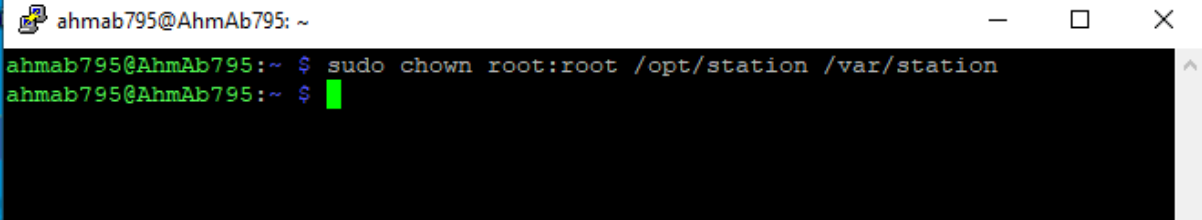
```
ahmab795@AhmAb795: ~
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
f3f27bca6588   hello-world    "/hello"                About a minute ago    Exited (0) About a
minute ago
200e3f3d6983   hello-world    "/hello"                2 minutes ago        Exited (0) 2 minute
s ago
ahmab795@AhmAb795:~$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS   NAMES
f3f27bca6588   hello-world    "/hello"                About a minute ago    Exited (0) About a minute ago
200e3f3d6983   hello-world    "/hello"                2 minutes ago        Exited (0) 2 minutes ago
ahmab795@AhmAb795:~$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS   NAMES
ahmab795@AhmAb795:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-13 10:35:34 GMT; 6min ago
     Invocation: 7e96c54a609e41f28ad87e63c8fb0ee5
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 13647 (dockerd)
      Tasks: 12
         CPU: 1.672s
        CGroup: /system.slice/docker.service
                └─13647 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Nov 13 10:35:34 AhmAb795 dockerd[13647]: time="2025-11-13T10:35:34.281590449Z" level=info msg="Docker daemon" commit=d105562 containerd-snapshotter=faib
Nov 13 10:35:34 AhmAb795 dockerd[13647]: time="2025-11-13T10:35:34.282307992Z" level=info msg="Initializing buildkit"
Nov 13 10:35:34 AhmAb795 dockerd[13647]: time="2025-11-13T10:35:34.345049834Z" level=info msg="Completed buildkit initialization"
Nov 13 10:35:34 AhmAb795 dockerd[13647]: time="2025-11-13T10:35:34.361069729Z" level=info msg="Daemon has completed initialization"
Nov 13 10:35:34 AhmAb795 dockerd[13647]: time="2025-11-13T10:35:34.361248577Z" level=info msg="API listen on /run/docker.sock"
Nov 13 10:35:34 AhmAb795 systemd[1]: Started docker.service - Docker Application Container Engine.
Nov 13 10:35:54 AhmAb795 dockerd[13647]: time="2025-11-13T10:35:54.792506758Z" level=info msg="sbJoin: gwep4 '-->'80cfe4936dbf', gwep6 '-->'eid=80cfe4936dbf'
Nov 13 10:35:54 AhmAb795 dockerd[13647]: time="2025-11-13T10:35:54.890729988Z" level=info msg="ignoring event" container=200e3f3d6983e0e7a2b5aa373a13c2b
Nov 13 10:37:28 AhmAb795 dockerd[13647]: time="2025-11-13T10:37:28.419163253Z" level=info msg="sbJoin: gwep4 '-->'452db297a2cd', gwep6 '-->'eid=452db297a2cd'
Nov 13 10:37:28 AhmAb795 dockerd[13647]: time="2025-11-13T10:37:28.504290271Z" level=info msg="ignoring event" container=f3f27bca6588e38fccb8780a0ee77b
lines 1-22/22 (END)
```

Creating required directories

Use the following commands to create the required directory in the root file system.

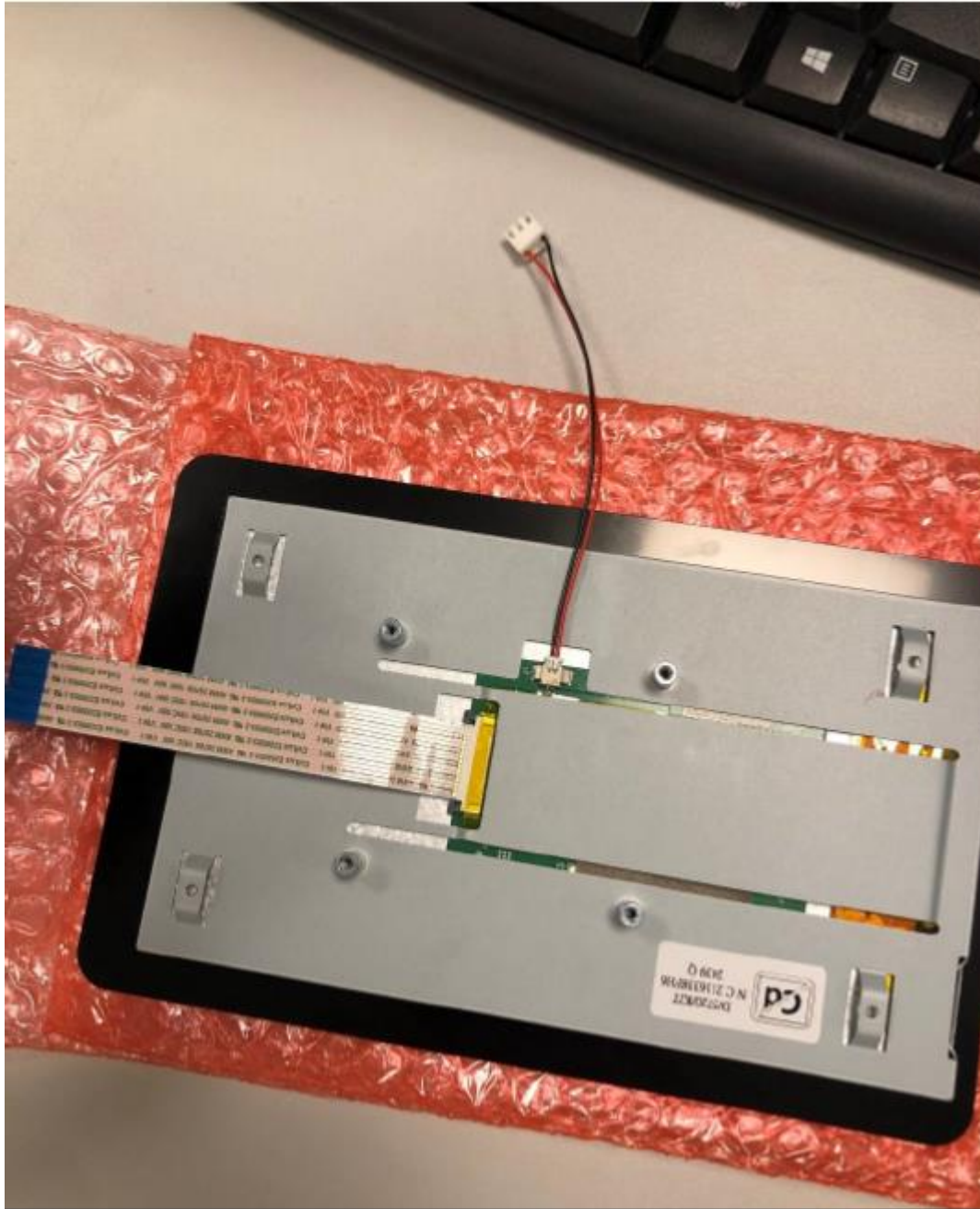
```
ahmab795@ahmab795:~$ sudo mkdir /opt/station
[sudo] password for ahmab795:
ahmab795@ahmab795:~$ sudo mkdir /var/station
ahmab795@ahmab795:~$
```



The image shows a terminal window with a title bar that reads 'ahmab795@AhmAb795: ~'. The terminal contains the same commands and output as the code block above, showing the successful creation of the /opt/station and /var/station directories.

Installation of screen

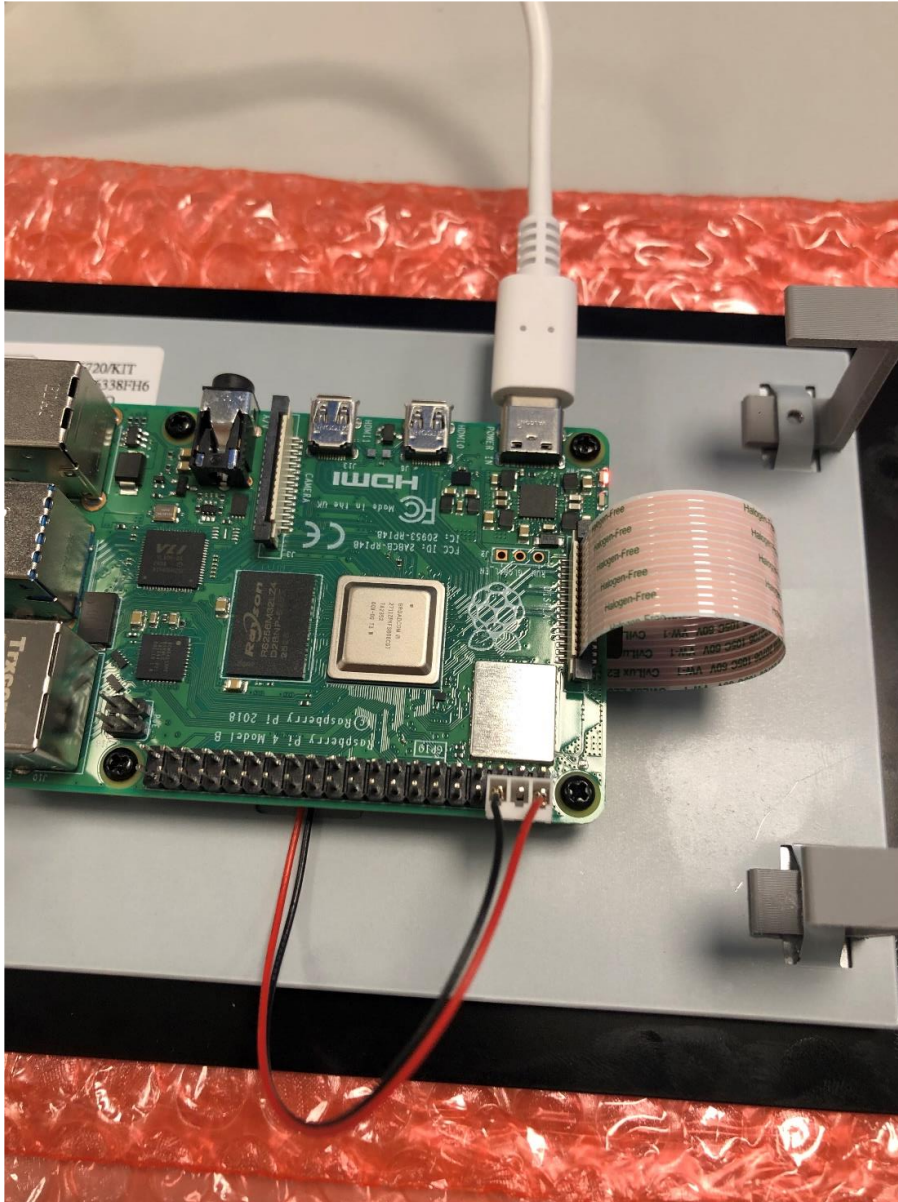
Connect the two provided video cables to the screen.



Place the Raspberry Pi next to the screen and connect the necessary cables.



Now, plug in the power cable and tighten the screws.



The holders shown below, which were 3D-printed by the teacher, allow us to flip the screen without damaging the Raspberry Pi behind it.

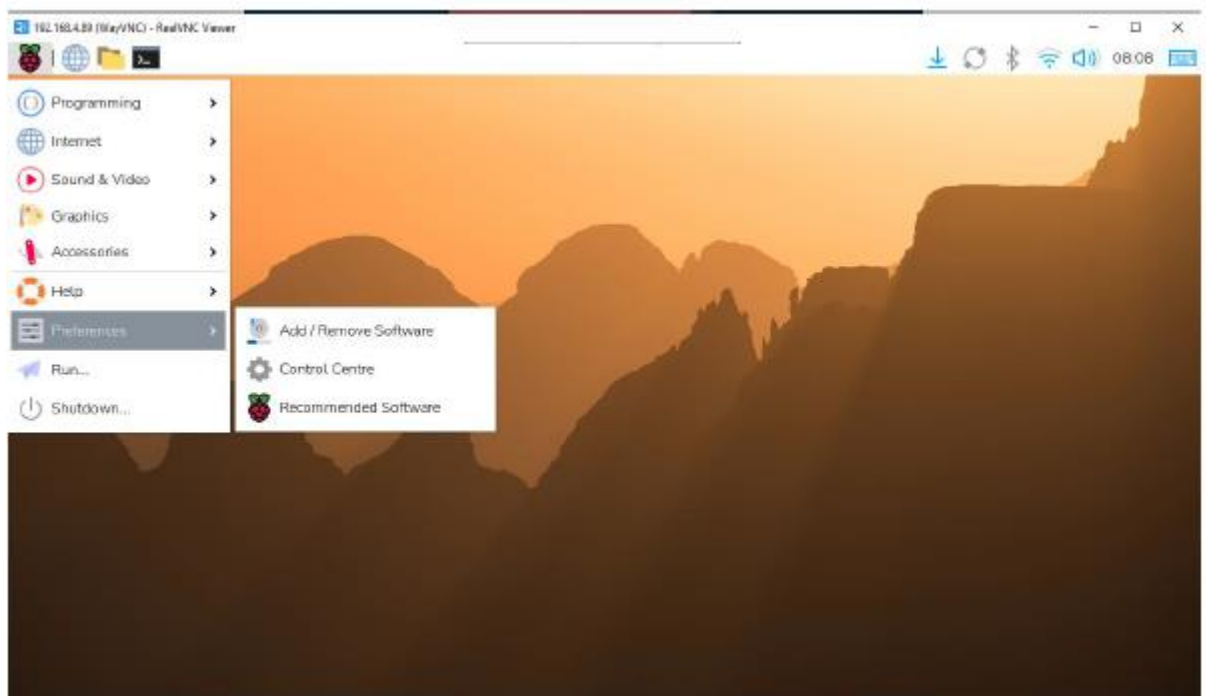


The screen orientation (landscape)

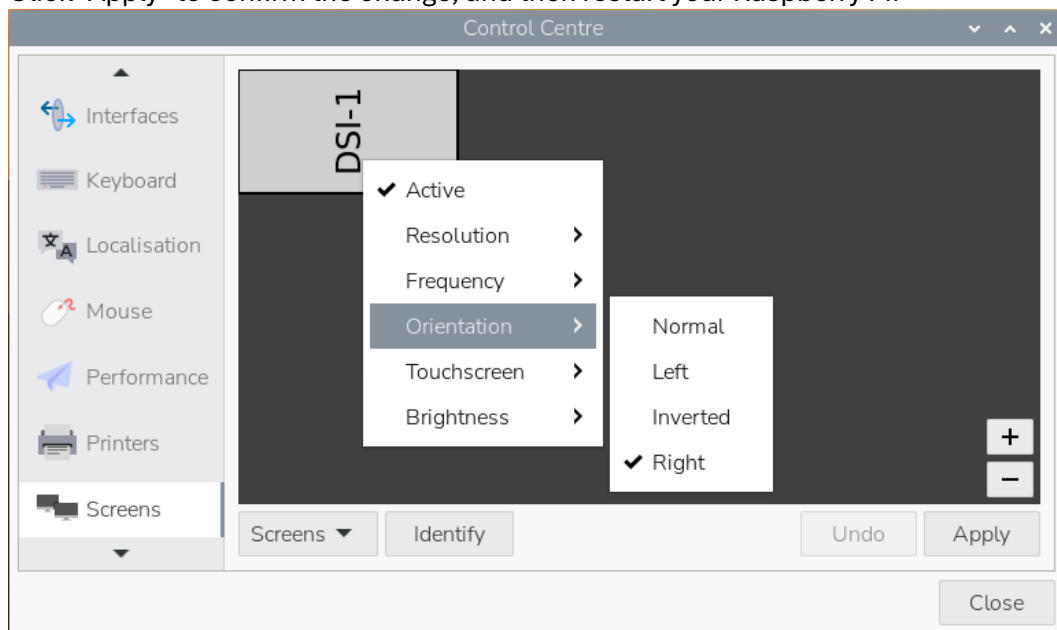
To change the Raspberry Pi's display orientation, follow these steps:

- Click the Raspberry Pi icon (Main Menu) in the top-left corner of the screen.

- Navigate to Preferences > Screen Configuration.



- Click on the Screens tab.
- Right-click on "DSI-1"
- Navigate to Orientation > Right.
- Click "Apply" to confirm the change, and then restart your Raspberry Pi.



Download Ressources and moving to right destination

The resource file has been uploaded to GitHub for convenient access. Use the following commands to clone the repository and access the "Ressourcen" folder.

```

ahmab795@AhmAb795: ~/Downloads
fatal: repository 'https://github.com/Abolooco/WEBAP/tree/main/Ressourcen/' not found
ahmab795@AhmAb795:~/Downloads $ sudo git clone https://github.com/Abolooco/WEBAP
Cloning into 'WEBAP'...
remote: Enumerating objects: 164, done.
remote: Counting objects: 100% (164/164), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 164 (delta 40), reused 150 (delta 26), pack-reused 0 (from 0)
Receiving objects: 100% (164/164), 1.60 MiB | 6.59 MiB/s, done.
Resolving deltas: 100% (40/40), done.
ahmab795@AhmAb795:~/Downloads $ ls
WEBAP
ahmab795@AhmAb795:~/Downloads $ cd WEBAP/
ahmab795@AhmAb795:~/Downloads/WEBAP $ ls
'Learning part' 'Porject Plan' Ressourcen 'Work Document'
ahmab795@AhmAb795:~/Downloads/WEBAP $ sudo mv -r Ressourcen /Download
mv: invalid option -- 'r'
Try 'mv --help' for more information.
ahmab795@AhmAb795:~/Downloads/WEBAP $ sudo mv -r Ressourcen /home/ahmab795/Downl
oads
mv: invalid option -- 'r'
Try 'mv --help' for more information.
ahmab795@AhmAb795:~/Downloads/WEBAP $ sudo mv Ressourcen /home/ahmab795/Download
s
ahmab795@AhmAb795:~/Downloads/WEBAP $ ls
'Learning part' 'Porject Plan' 'Work Document'
ahmab795@AhmAb795:~/Downloads/WEBAP $ cd ..
ahmab795@AhmAb795:~/Downloads $ ls
Ressourcen WEBAP
ahmab795@AhmAb795:~/Downloads $ sudo rm -d WEBAP/
rm: cannot remove 'WEBAP/': Directory not empty
ahmab795@AhmAb795:~/Downloads $ sudo rm -r WEBAP/
ahmab795@AhmAb795:~/Downloads $ ls
Ressourcen
ahmab795@AhmAb795:~/Downloads $

```

Move the firmware, configuration and scripts in "/opt/station" using commands below:

```

Sudo mkdir /opt/station/www
sudo mv ~/Downloads/Ressourcen/Station-Webseite/index.php /opt/station/www
sudo mv ~/Downloads/Ressourcen/'Firm und Scripts'/* /opt/station/

```

```

ahmab795@AhmAb795: ~
ahmab795@AhmAb795:~ $ ls /opt/station
bhl750  config.toml  mql35      test-bme280.py  www
bme280  measure.py   test-bhl750.py test-mql35.py
ahmab795@AhmAb795:~ $ ls /opt/station/www
index.php
ahmab795@AhmAb795:~ $

```

Create a Docker container (php:8.4-apache)

Use the provided command to create the Docker container

```

Sudo docker run -d --name php -p 8080:80 -v /tmp/station:/tmp/station -v
opt/station/www:/var/www/html php:8.4-apache

```

```
ahmab795@AhmAb795: ~  
ahmab795@AhmAb795:~$ sudo docker run -d --name php --restart unless-stopped -p 8080:80 -v /tmp/station:/tmp/statio  
n -v /opt/station/www:/var/www/html php:8.4-apache  
809dacd9127778d6851f60cf39ad0b8576a663673a455f55a16b3ba980dd6070  
ahmab795@AhmAb795:~$
```

Configure autostart

Since no actual sensor data or result.csv file exists initially, a Python script (measure.py) will generate a placeholder CSV file with simulated values for system testing.

This script will also automate the following startup procedures:

1. **Generate Test Data:** Create a result.csv file containing fake measurement values if it does not already exist.
2. **Deploy Application:** Run the Docker container to host the web application.
3. **Expose Web Interface:** Make the website available at <http://localhost:8080>.
4. **Configure Kiosk Mode:** Launch the browser in a kiosk mode with a blank password for unattended operation.
5. **Display Data:** Ensure the current (simulated) measured values are displayed on the webpage.

This entire process is configured to execute automatically upon system boot.

```
ahmab795@AhmAb795:~$ sudo nano ~/.config/labwc/autostart  
ahmab795@AhmAb795:~$ sudo chmod +x ~/.config/labwc/autostart  
ahmab795@AhmAb795:~$
```

```

ahmab795@AhmAb795: ~
GNU nano 8.4 /home/ahmab795/.config/labwc/autostart
#!/bin/bash

mkdir -p /tmp/station
python3 /opt/station/measure.py --offline > /tmp/station/result.csv &
docker start php &
chromium --kiosk http://localhost:8080 --password-store=basic &
  
```

[Wrote 6 lines]

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Time	2025-11-20 15:28:42
Temperature	20.00 °C
Humidity	50.00 %
Pressure	1,000.00 hPa
Light	0.00 lux
Gas	500.00 ppm

Database

Introduction

The database system is designed to store, manage, and organize measurement data recorded by weather monitoring stations. The main goals of the design are:

- To store user accounts, their stations, and recorded measurements
- To allow users to organize measurements into collections
- To support sharing collections with other users
- To support friendships between users
- To enable administrators to create stations before they are assigned to customers

The database model is based on a Conceptual Data Model (MCD) and its corresponding Logical Data Model (LDM/MLD).

Website

A simple web designed to satisfy the requirements. This website has the following features:

- A user can be created and login
- Users can manage their credentials
- Existing stations could be assigned to users
- A friendship request can be made and target user may accept or reject it
- Creating collection of measurement is possible
- Collection of measurements can be shared with friends

Admin panel:

- View total number of users, station, measurements and collections
- Creating new users
- View all users, change privileges and delete users
- Creating new station
- Edit information of station
- View all station's measurements and collections

Web/Database server

Introduction

This part describes the installation and configuration of a server that functions as both a web server and a database server. The purpose of this project is to set up a working environment in which a website and a database can operate together on the same system.

The server is configured with **Apache HTTP** Server as the web server, **MySQL** as the database server, and PHP to process dynamic web content. Additionally, **phpMyAdmin** is installed to allow graphical database administration through a web browser.

The objective of this work is to:

- Install and configure all required software components
- Create a test database and a test website including test users
- Ensure that the website is accessible via a web browser
- Apply basic security measures to protect the system

The installation of the Ubuntu operating system itself is not part of this documentation. Instead, the focus lies on the web and database services, their configuration, security settings, and verification of functionality.

Implementation Steps

Following the user story we can progress step-by-step with the following topics.

- Installation of requirements
- Create a PHP webpage and database
- Test your PHP file and DB
- Connection to DB

Installation of requirements

- **Update and upgrade package list**

```
ahmab795@pif2026:~$ sudo apt update && sudo apt upgrade -y
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:1 https://lu.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 https://lu.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 https://lu.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

- **Install Apache**

Sudo apt install apache2

```
ahmab795@pif2026:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-03-04 07:48:59 UTC; 4min 9s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 17507 (apache2)
      Tasks: 55 (limit: 13305)
     Memory: 5.4M (peak: 6.2M)
        CPU: 75ms
    CGroup: /system.slice/apache2.service
            └─17507 /usr/sbin/apache2 -k start
              └─17508 /usr/sbin/apache2 -k start
                └─17510 /usr/sbin/apache2 -k start
```

- **Install PHP**

Sudo apt install php

- **Install MySQL**

Sudo apt install php libapache2-mod-php

```
ahmab795@pif2026:~$ sudo apt install php libapache2-mod-php
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
php is already the newest version (2:8.3+93ubuntu2).
libapache2-mod-php is already the newest version (2:8.3+93ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

- **Install MySQLi extension**

sudo apt install php-mysql

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  php8.3-mysql
The following NEW packages will be installed:
  php-mysql php8.3-mysql
0 upgraded, 2 newly installed, 0 to remove and 1 not upgraded.
Need to get 129 kB of archives.
After this operation, 473 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 https://lu.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-mysql amd64 8.3.6-0ubuntu2
Get:2 http://lu.archive.ubuntu.com/ubuntu noble/main amd64 php-mysql all 2:8.3+93ubuntu2 [1,838 B]
```

- **Install SSH**

Sudo apt install openssh-server

```

ahmab795@pif2026:/var/www/html$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.6p1-3ubuntu13.14).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ahmab795@pif2026:/var/www/html$

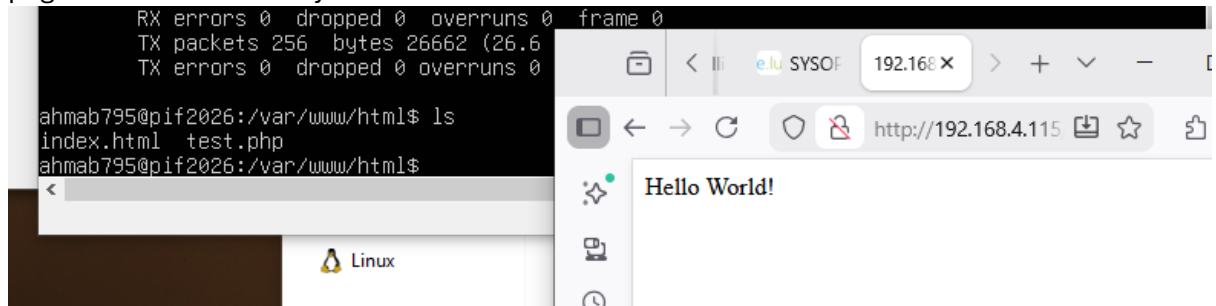
```

- Once installation completed you can restart the apache2

```
ahmab795@pif2026:~$ sudo systemctl restart apache2
```

- **Test PHP for apache2**

page loaded successfully



Create test Database

Login to MySQL and change your password with “mysql_secure_installation”.

```

Mar 08 18:48:23 pif2026 systemd[1]: Started MySQL Database Server.
ahmab795@pif2026:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with \n.
MySQL> alter user 'root'@'localhost' IDENTIFIED with mysql_native_password by 'mysql_secure_installation'
-> ;
Query OK, 0 rows affected (0.09 sec)
mysql> use 1TPIFI
mysql> e changed

```

Create a database for example called “1TPIFI” and switch to it.

Create database “1TPIFI”;
Use 1TPIFI;

Create a test table called “student” and insert some value to it

```
mysql> create table student(
  -> ID INT primary key,
  -> Name Varchar(60),
  -> Age INT
  -> );
Query OK, 0 rows affected (0.25 sec)

mysql>

mysql> insert into student(ID, Name, Age) value (1, 'Abolo_2', 23);
ERROR 1062 (23000): Duplicate entry '1' for key 'student.PRIMARY'
mysql> insert into student(ID, Name, Age) value (2, 'Abolo_2', 23);
Query OK, 1 row affected (0.07 sec)

mysql> insert into student(ID, Name, Age) value (3, 'Abolo_3', 23);
Query OK, 1 row affected (0.04 sec)

mysql> insert into student(ID, Name, Age) value (4, 'Abolo_4', 24);
Query OK, 1 row affected (0.14 sec)

mysql>
```

Use keyword “exit” to logout and login using directly to your table shown below

```
ahmab795@pif2026:~$ sudo mysql -p 1TPIFI
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 21
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

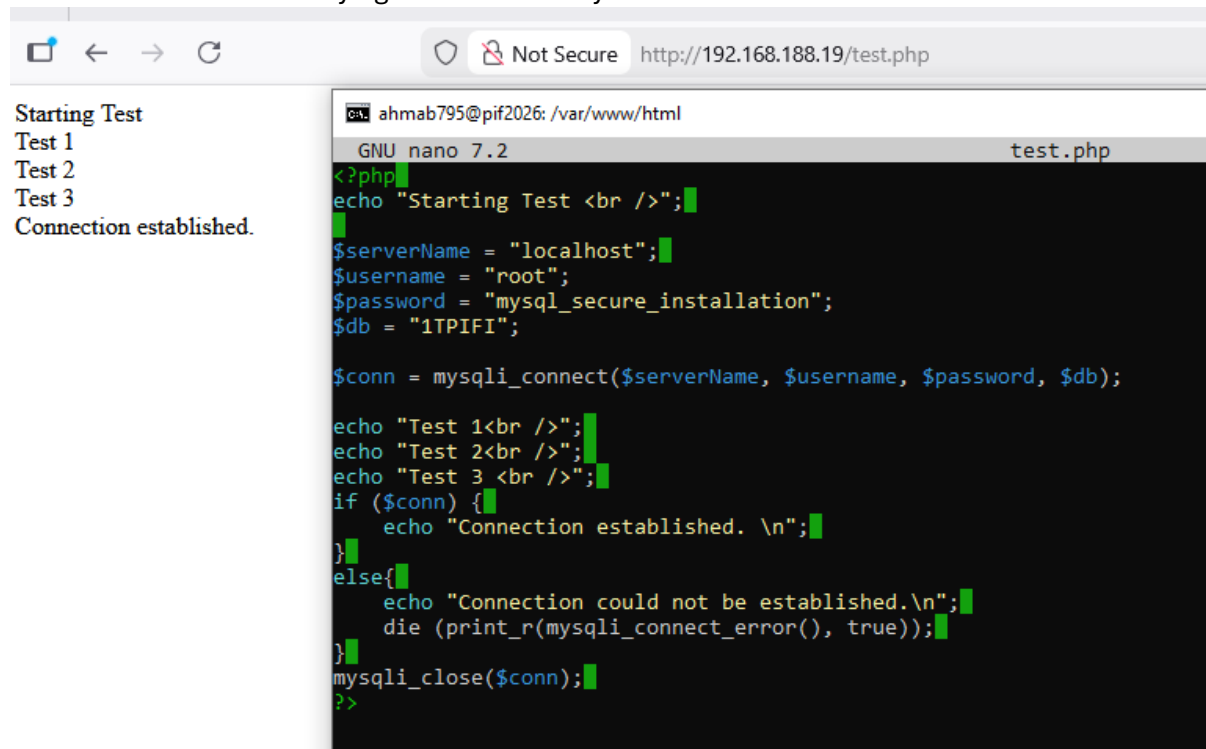
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select * from student;
+-----+-----+-----+
| ID | Name   | Age |
+-----+-----+-----+
| 1  | Abolo  | 22  |
| 2  | Abolo_2 | 23  |
| 3  | Abolo_3 | 23  |
| 4  | Abolo_4 | 24  |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

Create test webpage

A test PHP were created trying to connect to my database.



DB interaction with webpage

The test.php file were developed to access “student” table specifically and display its content. An unexpected error happened which was a bit challenging for me to fix because as we can see there is no error message as clue.

Looks like there’s a problem with this site

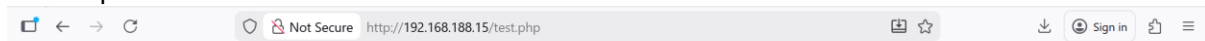
http://192.168.188.15/test.php might have a temporary problem or it could have moved.

Error code: 500 Internal Server Error

- The site could be temporarily unavailable or too busy. Try again in a few moments.

Try Again

Fortunality, the problem was not so difficult to fix since small SQL syntax issue were causing me to spend a bit more time on it.



Starting Test
Test 1
Test 2
Test 3

Display table: 1TPIFI

ID	Name	Age
1	Abolo	22
2	Abolo_2	23
3	Abolo_3	23
4	Abolo_4	24

Here is the confirmation of DB connection.

Restrictions and permissions

Setting all files to 640 (owner read/write, group read, others NO access)

```
ahmab795@pif2026:/var/www/html$ sudo find /var/www/html/ -type f -exec chmod 640 {} \;
```

DB restricted user

Created user with "website_user" username and assigned a password for it. The minimum permission and accessibility has been granted.

```
mysql> grant select,insert,update,delete on PIF_2026.* to 'website_user'@'localhost';
Query OK, 0 rows affected (0.14 sec)

mysql> flush privileges;
Query OK, 0 rows affected (0.07 sec)
```

Verification of Privileges

```
mysql> show grants for 'website_user'@'localhost';
+-----+
| Grants for website_user@localhost |
+-----+
| GRANT USAGE ON *.* TO 'website_user'@'localhost' |
| GRANT SELECT, INSERT, UPDATE, DELETE ON `PIF_2026`.* TO 'website_user'@'localhost' |
+-----+
2 rows in set (0.01 sec)
```

PhpMyAdmin

- Download the file from official website

```
ahmab795@pif2026:/var/www/html$ sudo wget https://files.phpmyadmin.net/phpMyAdmin/5.2.2/phpMyAdmin-5.2.2-all-languages.zip
--2026-03-11 20:24:07-- https://files.phpmyadmin.net/phpMyAdmin/5.2.2/phpMyAdmin-5.2.2-all-languages.zip
Resolving files.phpmyadmin.net (files.phpmyadmin.net)... 207.211.214.80, 2a02:6ea0:fa77::3
Connecting to files.phpmyadmin.net (files.phpmyadmin.net)|207.211.214.80|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 15782614 (15M) [application/zip]
Saving to: 'phpMyAdmin-5.2.2-all-languages.zip'

phpMyAdmin-5.2.2-all-language 100%[=====] 15.05M 9.21MB/s in 1.6s
2026-03-11 20:24:09 (9.21 MB/s) - 'phpMyAdmin-5.2.2-all-languages.zip' saved [15782614/15782614]
```

- Once the download is finished you can extract the files into /var/www/html

```

ahmab795@pif2026:/var/www/html$ sudo apt install unzip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unzip is already the newest version.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

ahmab795@pif2026:/var/www/html$ sudo unzip phpMyAdmin-5.2.3-all-languages.zip

ahmab795@pif2026:/var/www/html$ ls
apache-default.html  img          Page
Database            js           phpMyAdmin-5.2.3-all-languages
DataHandler.php     MyLibrary.php phpMyAdmin-5.2.3-all-languages.zip
DataSenders         MyStyle.css
db_config.php       note
ahmab795@pif2026:/var/www/html$ sudo mv phpMyAdmin-5.2.3-all-languages phpmyadmin
ahmab795@pif2026:/var/www/html$ ls
apache-default.html  db_config.php  MyStyle.css  phpMyAdmin-5.2.3-all-languages.zip
Database            img           note
DataHandler.php     js           Page
DataSenders         MyLibrary.php phpmyadmin
ahmab795@pif2026:/var/www/html$

```

- Navigate to the phpmyadmin, phpMyAdmin comes with a sample configuration file. You need to copy it to create the main configuration file.

```

ahmab795@pif2026:/var/www/html$ cd phpmyadmin/
ahmab795@pif2026:/var/www/html/phpmyadmin$ ls
babel.config.json  config.sample.inc.php  favicon.ico  LICENSE  RELEASE-DATE-5.2.2  sql  vendor
ChangeLog          CONTRIBUTING.md        index.php    locale  robots.txt          templates  yarn.lock
composer.json      doc                   js          package.json  setup              themes
composer.lock      examples              libraries    README      show_config_errors.php  url.php
ahmab795@pif2026:/var/www/html/phpmyadmin$ sudo cp config.sample.inc.php config.inc.php
ahmab795@pif2026:/var/www/html/phpmyadmin$

```

- Start editing “config.inc.php” using nano text editor or any other text editor
- set the blowfish_secret: Find the line \$cfg['blowfish_secret'] = '';. You must fill this with a random 32-character string. This is used for cookie encryption. You can generate one online or just type a random string of letters and numbers. It should look something like this:

```

ahmab795@pif2026: /var/www/html/phpmyadmin
GNU nano 7.2                                config.inc.php *
<?php
/**
 * phpMyAdmin sample configuration, you can use it as base for
 * manual configuration. For easier setup you can use setup/
 *
 * All directives are explained in documentation in the doc/ folder
 * or at <https://docs.phpmyadmin.net/>.
 */

declare(strict_types=1);

/**
 * This is needed for cookie based authentication to encrypt the cookie.
 * Needs to be a 32-bytes long string of random bytes. See FAQ 2.10.
 */
$cfg['blowfish_secret'] = 'aB3dE5fG7hIjKlMnOpQrStUvWxYz0123'; /* YOU MUST FILL IN THIS FOR COOKIE AUTH! */

/**
 * Servers configuration
 */
$i = 0;

/**

```

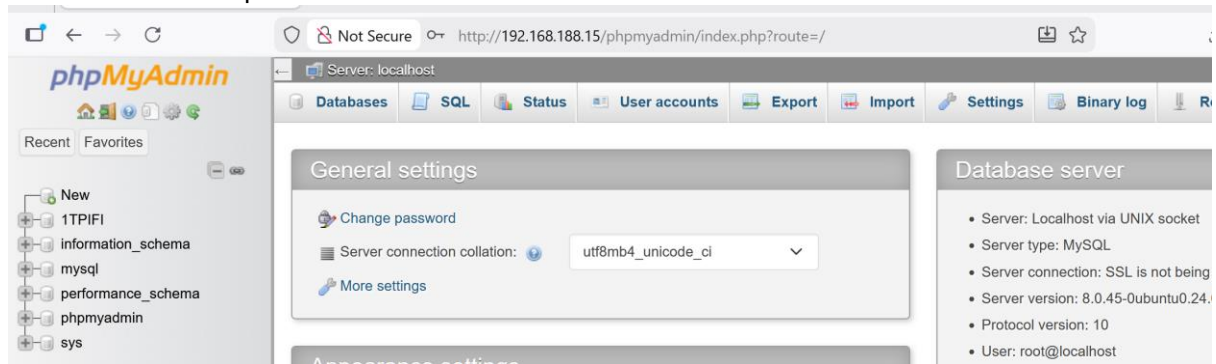
- set proper permissions

```

ahmab795@pif2026:/var/www/html/phpmyadmin$ sudo chown -R ahmab795:www-data /var/www/html/phpmyadmin

```

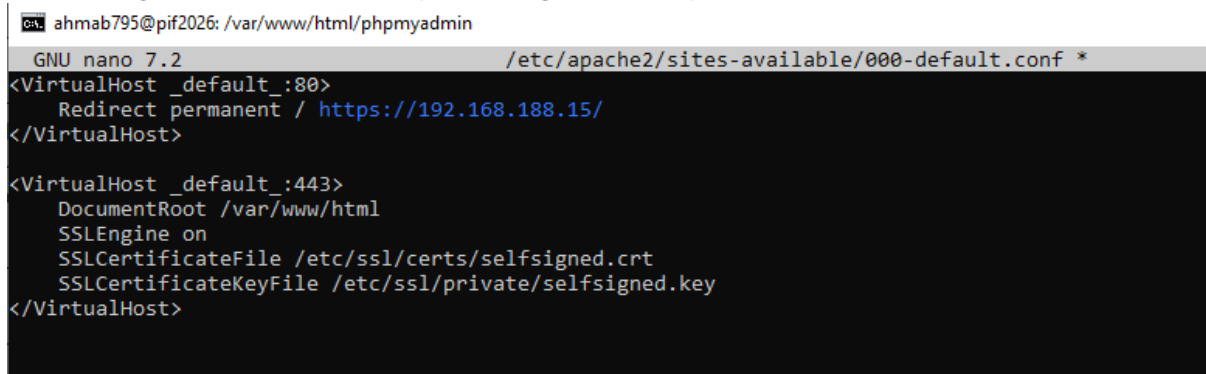

- Verification of setup



I used my mysql credentials(root as username and mysql_secure_installation as password) to login and access to phpmyadmin.

HTTPS redirection

This setting catches all traffic on port 80 regardless of ip or hostname



SSL self-signed certificate

Since we are using an IP address, you need a self-signed certificate:

- Create SSL directory

```
ahmab795@pif2026:/var/www/html/phpmyadmin$ sudo mkdir -p /etc/ssl/private
ahmab795@pif2026:/var/www/html/phpmyadmin$ sudo mkdir -p /etc/ssl/certs
```

- Generate self-signed certificate for IP address

```
-ahmab795@pif2026:/var/www/html/phpmyadmin$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
> -subj "/C=US/ST=State/L=City/O=Organization/CN=192.168.188.15" \
> -keyout /etc/ssl/private/selfsigned.key \
> -out /etc/ssl/certs/selfsigned.crt
```

- set proper permissions

```
ahmab795@pif2026:/var/www/html/phpmyadmin$ sudo chmod 600 /etc/ssl/private/selfsigned.key
```

- Enable Required Modules

```
ahmab795@pif2026:/var/www/html/phpmyadmin$ sudo a2enmod ssl
sudo a2enmod rewrite
sudo systemctl restart apache2Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
systemctl restart apache2
```

- Configure Virtual Host

```
ahmab795@pif2026: /var/www/html/phpmyadmin
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf *

<VirtualHost *:80>
    # Redirect all HTTP traffic to HTTPS
    RewriteEngine On
    RewriteCond %{HTTPS} off
    RewriteRule ^(.*)$ https://%{HTTP_HOST}/$1 [R=301,L]

    # Alternative simple redirect
    # Redirect permanent / https://192.168.188.15/
</VirtualHost>

<VirtualHost *:443>
    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html

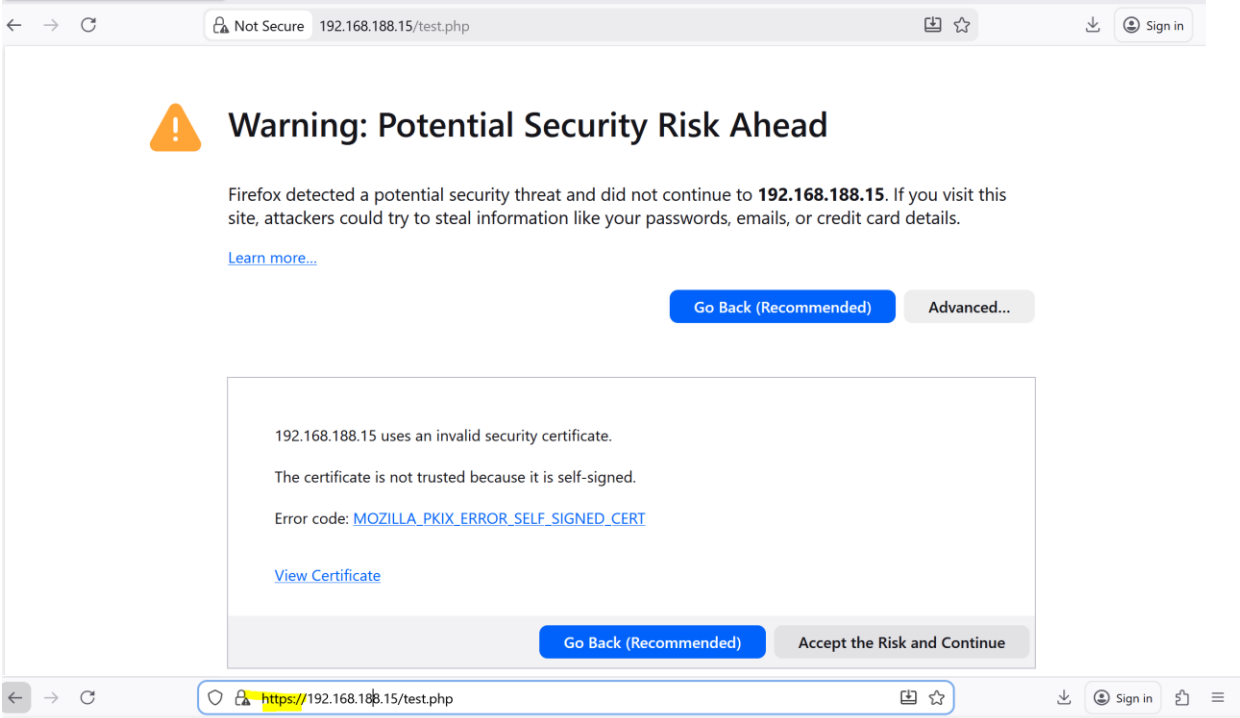
    # Log files
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # SSL Configuration
    SSLEngine on
    SSLCertificateFile /etc/ssl/certs/selfsigned.crt
    SSLCertificateKeyFile /etc/ssl/private/selfsigned.key

    # Security Headers (optional)
    Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"

    <Directory /var/www/html>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

- Test and Reload



Starting Test
Test 1
Test 2
Test 3

Display table: 1TPIFI

ID	Name	Age
1	Abolo	22
2	Abolo 2	23
3	Abolo 3	23
4	Abolo 4	24

Here is the confirmation of DB connection.

Data transfer

Address initialization

Initialize your webserver address.

```
ahmab795@pif2026:/var/www/html$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.4.157 netmask 255.255.254.0 broadcast 192.168.5.255
    inet6 fe80::a00:27ff:fe43:e561 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:43:e5:61 txqueuelen 1000 (Ethernet)
    RX packets 37273 bytes 15643275 (15.6 MB)
    RX errors 0 dropped 1908 overruns 0 frame 0
    TX packets 24593 bytes 2398078 (2.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

IP: 192.168.4.157

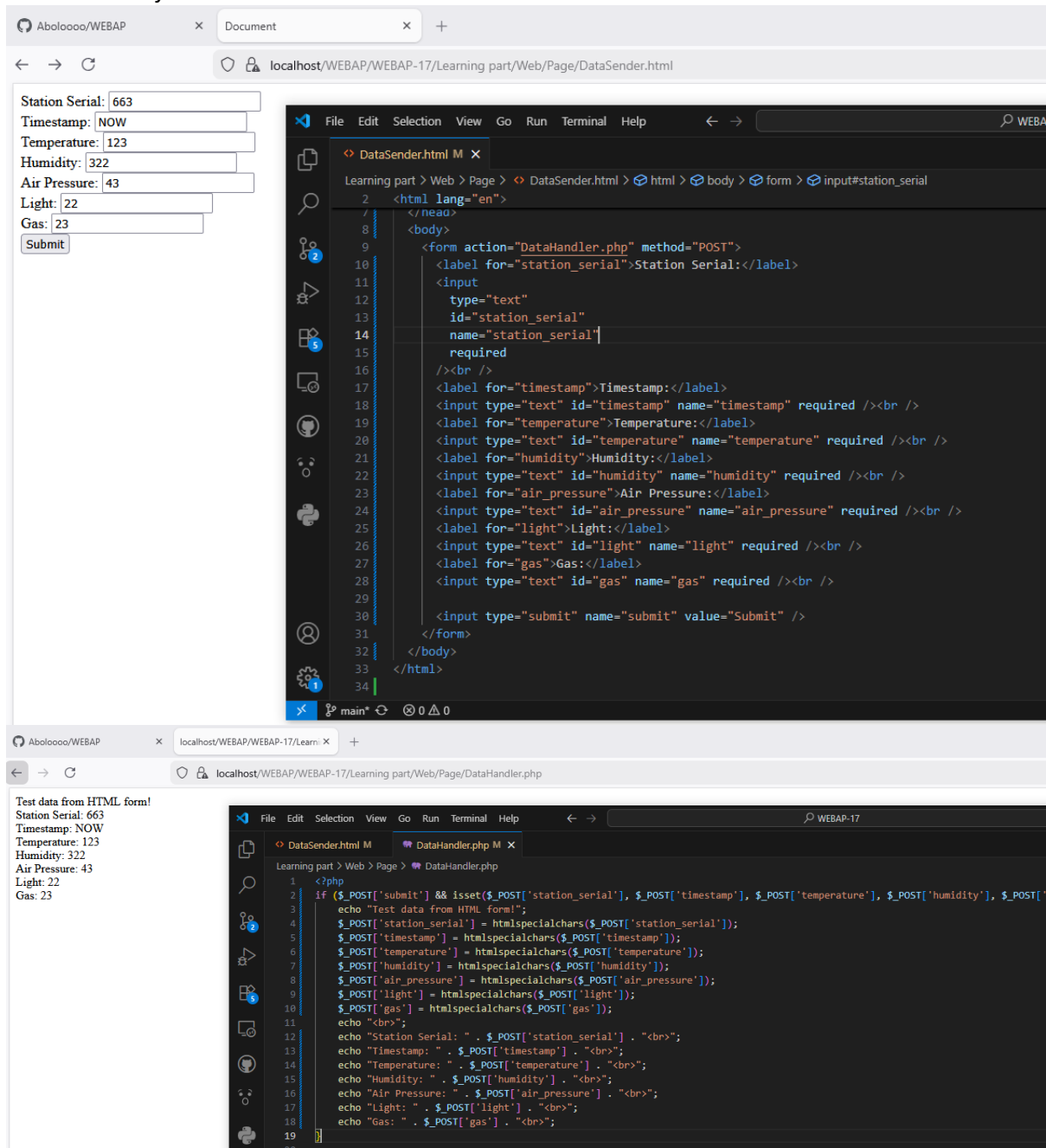
Path: /DataHandler.php

File structure

/var/www/html -> MyLibrary.php | MyScript.js | MyStyle.css -> /Web -> index.php

Test Data Transfer

Create two .php and html files. One for creating new test values and one for handling them. The point of transferring test values is to make sure values can be sent through files since the actual values from Python wouldn't be visible.



The screenshot shows a web browser window with a form for entering test data. The form fields are: Station Serial: 663, Timestamp: NOW, Temperature: 123, Humidity: 322, Air Pressure: 43, Light: 22, Gas: 23, and a Submit button. The browser address bar shows the URL: localhost/WEBAP/WEBAP-17/Learning part/Web/Page/DataSender.html.

Below the browser window, the code editor shows the HTML file (DataSender.html) and the PHP file (DataHandler.php). The HTML file contains a form with the following structure:

```

<html lang="en">
</head>
<body>
<form action="DataHandler.php" method="POST">
<label for="station_serial">Station Serial:</label>
<input type="text" id="station_serial" name="station_serial" required />
<br />
<label for="timestamp">Timestamp:</label>
<input type="text" id="timestamp" name="timestamp" required />
<br />
<label for="temperature">Temperature:</label>
<input type="text" id="temperature" name="temperature" required />
<br />
<label for="humidity">Humidity:</label>
<input type="text" id="humidity" name="humidity" required />
<br />
<label for="air_pressure">Air Pressure:</label>
<input type="text" id="air_pressure" name="air_pressure" required />
<br />
<label for="light">Light:</label>
<input type="text" id="light" name="light" required />
<br />
<label for="gas">Gas:</label>
<input type="text" id="gas" name="gas" required />
<br />
<input type="submit" name="submit" value="Submit" />
</form>
</body>
</html>

```

The PHP file (DataHandler.php) contains the following code:

```

<?php
if ($_POST['submit'] && isset($_POST['station_serial'], $_POST['timestamp'], $_POST['temperature'], $_POST['humidity'], $_POST['air_pressure'], $_POST['light'], $_POST['gas'])) {
    echo "Test data from HTML form!";
    $_POST['station_serial'] = htmlspecialchars($_POST['station_serial']);
    $_POST['timestamp'] = htmlspecialchars($_POST['timestamp']);
    $_POST['temperature'] = htmlspecialchars($_POST['temperature']);
    $_POST['humidity'] = htmlspecialchars($_POST['humidity']);
    $_POST['air_pressure'] = htmlspecialchars($_POST['air_pressure']);
    $_POST['light'] = htmlspecialchars($_POST['light']);
    $_POST['gas'] = htmlspecialchars($_POST['gas']);
    echo "<br>";
    echo "Station Serial: " . $_POST['station_serial'] . "<br>";
    echo "Timestamp: " . $_POST['timestamp'] . "<br>";
    echo "Temperature: " . $_POST['temperature'] . "<br>";
    echo "Humidity: " . $_POST['humidity'] . "<br>";
    echo "Air Pressure: " . $_POST['air_pressure'] . "<br>";
    echo "Light: " . $_POST['light'] . "<br>";
    echo "Gas: " . $_POST['gas'] . "<br>";
}

```

Data transered successfully.

Final script

The following php script will help us to:

- Receive values from station
- Make connection to database
- Insert values into measurement table

```

1  <?php
2  // Check if all required POST parameters exist
3  if (
4      isset(
5          $_POST['station_serial'],
6          $_POST['timestamp'],
7          $_POST['temperature'],
8          $_POST['humidity'],
9          $_POST['pressure'],
10         $_POST['light'],
11         $_POST['gas']
12     )
13 ) {
14     $station_serial = $_POST['station_serial'];
15     $timestamp       = $_POST['timestamp'];
16     $temperature     = (float)$_POST['temperature'];
17     $humidity        = (float)$_POST['humidity'];
18     $pressure        = (float)$_POST['pressure'];
19     $light           = (float)$_POST['light'];
20     $air_quality     = (int)$_POST['gas'];
21 } else {
22     die("Error: Missing required POST parameters.");
23 }

25 $host = 'localhost';
26 $username = 'root';
27 $password = 'mysql_secure_installation';
28 $database = 'PIF_2026';
29
30 $connection = mysqli_connect($host, $username, $password, $database);
31
32 if (!$connection) {
33     die("Connection failed: " . mysqli_connect_error());
34 }
35
36 mysqli_set_charset($connection, "utf8mb4");
37
38 if (strpos($timestamp, '.') !== false) {
39     $timestamp = explode('.', $timestamp)[0];
40 }
41
42 $station_query = "SELECT Station_id FROM Station WHERE Serial_number = ?";
43 $stmt = $connection->prepare($station_query);
44 $stmt->bind_param("s", $station_serial);
45 $stmt->execute();
46 $result = $stmt->get_result();
47
48 if ($row = $result->fetch_assoc()) {
49     $station_id = $row['Station_id'];
50     $stmt->close();
51
52     $insert_query = "INSERT INTO Measurement (Timestamp, Temperature, Humidity, Air_pressure, Light_intensity, Air_quality, Station_id)
53                     VALUES (?, ?, ?, ?, ?, ?, ?)";
54
55     $stat = $connection->prepare($insert_query);
56
57     $stat->bind_param("sddddii", $timestamp, $temperature, $humidity, $pressure, $light, $air_quality, $station_id);
58
59     if ($stat->execute()) {
60         echo "Data inserted successfully. Measurement ID: " . $stat->insert_id;
61     } else {
62         echo "Error inserting data: " . $stat->error;
63     }
64     $stat->close();
65 } else {
66     echo "Error: Station with Serial_number '$station_serial' not found in Station table. Please add it first.";
67     $stmt->close();
68 }
69
70 $connection->close();
71 ?>

```

Station configuration

Start your station configuration by editing “/opt/station/config.toml”, here we can initialize our station:

- serial number
- IP address
- Path


```

File Edit Tabs Help
GNU nano 8.4 config.toml
station_serial = "WST-202601-001" # Serial number of the stati>
interval = 2 # Time between measurements
toggle_intensity = 100 # The light level at which the LED should b>
[i2c] # I2C-Addresses of the sensors
bme280 = 0x76
bh1750 = 0x23
mq135 = 0x48
[destination] # Information about the server
ip = "192.168.4.157" # IP-Address or Hostname of the Server
path = "/DataHandler.php" # Path to the script that accepts the measu>

[ Wrote 14 lines ]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

```

Run “measure.py” to start your station measuring. What measure.py does is

- Start Getting sensors values
- Send the values to webserver to specific path(DataHandler.php)

Use Wireshark to see if the data are being send to target destination(webserver) properly.

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

(ip.addr == 192.168.4.141) && (_ws.col.protocol == "HTTP")

No.	Time	Source	Destination	Protocol	Length	Info
1478	12.886833	192.168.4.141	192.168.4.66	HTTP	217	POST /WEBAP/WEBAP-17/Learning%20part/Web/Page/DataHandler.php HTTP/1.1
1480	12.888218	192.168.4.66	192.168.4.141	HTTP	306	HTTP/1.1 200 OK
1599	14.917243	192.168.4.141	192.168.4.66	HTTP	203	POST /WEBAP/WEBAP-17/Learning%20part/Web/Page/DataHandler.php HTTP/1.1
1601	14.921165	192.168.4.66	192.168.4.141	HTTP	306	HTTP/1.1 200 OK
1678	16.950308	192.168.4.141	192.168.4.66	HTTP	218	POST /WEBAP/WEBAP-17/Learning%20part/Web/Page/DataHandler.php HTTP/1.1
1680	16.954234	192.168.4.66	192.168.4.141	HTTP	306	HTTP/1.1 200 OK
1753	18.990427	192.168.4.141	192.168.4.66	HTTP	203	POST /WEBAP/WEBAP-17/Learning%20part/Web/Page/DataHandler.php HTTP/1.1

Frame 1599: 203 bytes on wire (1624 bits), 203 bytes captured (1624 bits) on interface \Device\NPF_{CFD5E460-187C-4EFF-B0EE-75B44660E} 0000 f0

Ethernet II, Src: RaspberryPi_59:c9:a3 (88:a2:9e:59:c9:a3), Dst: ASUSTekCOMPU_82:58:77 (f0:2f:74:82:58:77) 0010 00

Internet Protocol Version 4, Src: 192.168.4.141, Dst: 192.168.4.66 0020 04

Transmission Control Protocol, Src Port: 43160, Dst Port: 80, Seq: 270, Ack: 1, Len: 149 0030 01

[2 Reassembled TCP Segments (418 bytes): #1598(269), #1599(149)] 0040 72

Hypertext Transfer Protocol 0050 6d

HTML Form URL Encoded: application/x-www-form-urlencoded 0060 25

Form item: "station_serial" = "102" 0070 30

Form item: "timestamp" = "2026-03-19 00:51:06.586720" 0080 2e

Form item: "temperature" = "26.263024026739004" 0090 26

Form item: "humidity" = "0.0" 00a0 65

Form item: "pressure" = "997.9721562289546" 00b0 36

Form item: "light" = "2.5" 00c0 2e

Form item: "gas" = "2005"

It is always good to double check if those measurements are adopted to your measurement table in your database.

Create Database

Login to webserver and mysql

```
ahmab795@pif2026:~$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.
```

Create your database according to your webpages configuration

```
mysql> create database PIF_2026;
Query OK, 1 row affected (0.11 sec)

mysql> use PIF_2026;
Database changed
mysql>
```

Importe sql file to the database PIF_2026

```
mysql> source /var/www/DB.sql;
Query OK, 0 rows affected (0.09 sec)

Query OK, 1 row affected (0.06 sec)

Database changed
Query OK, 0 rows affected (0.40 sec)

Query OK, 1 row affected (0.13 sec)
```

Importing of sql file done successful.

```
mysql> select * from Measurement;
+-----+-----+-----+-----+-----+-----+-----+-----+
| Measurement_id | Timestamp           | Temperature | Humidity | Air_pressure | Light_intensity | Air_quality | Station |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | 2026-01-18 08:00:00 | 22.0000 | 45.0000 | 1013.0000 | 350.0000 | 1 | 1 |
| 2 | 2026-01-18 10:00:00 | 24.0000 | 42.0000 | 1012.0000 | 850.0000 | 1 | 1 |
+-----+-----+-----+-----+-----+-----+-----+-----+
78 rows in set (0.00 sec)
```

Move all the webpages and assistance to “/var/www/html” and “/var/www/html/Web”

```
ahmab795@pif2026:/var/www/html$ ls
DataHandler.php  index-test.html  MyScript.js  phpmyadmin  test.php  WEBAP
debug.log       MyLibrary.php   MyStyle.css  phpMyAdmin-5.2.2-all-languages.zip  Web
ahmab795@pif2026:/var/www/html$ cd Web/
ahmab795@pif2026:/var/www/html/Web$ ls
admin.php      DataSender.html  Friendship.php  sign_in_up.php
Collection.php  debug.log        index.php       StationRegistration.php
ahmab795@pif2026:/var/www/html/Web$
```

Verify insert of new measurement

Check your measurement table for the newest values

65	2026-03-20 03:53:54	26.2147	0.0000	991.8484	1.6667	1875	1
66	2026-03-20 03:53:57	26.2128	0.0000	991.8535	1.6667	1875	1
67	2026-03-20 03:53:59	26.2080	0.0000	991.8505	1.6667	1876	1
68	2026-03-20 03:54:01	26.2042	0.0000	991.8574	1.6667	1876	1
69	2026-03-20 03:54:03	26.1994	0.0000	991.8594	1.6667	1875	1
70	2026-03-20 03:54:05	26.1934	0.0000	991.8559	1.6667	1875	1
71	2026-03-20 03:54:07	26.1870	0.0000	991.8335	1.6667	1878	1
72	2026-03-20 03:54:09	26.1838	0.0000	991.8415	1.6667	1878	1
73	2026-03-20 03:54:11	26.1838	0.0000	991.8381	1.6667	1877	1
74	2026-03-20 03:54:13	26.1823	0.0000	991.8371	1.6667	1874	1
75	2026-03-20 03:54:15	26.1797	0.0000	991.8262	1.6667	1874	1
76	2026-03-20 03:54:17	26.1819	0.0000	991.8299	1.6667	1875	1
77	2026-03-20 03:54:19	26.1819	0.0000	991.8349	1.6667	1876	1
78	2026-03-20 03:54:21	26.1816	0.0000	991.8260	1.6667	1876	1
79	2026-03-20 03:54:24	26.1797	0.0000	991.8295	1.6667	1877	1
80	2026-03-20 03:54:26	26.1778	0.0000	991.8330	1.6667	1876	1
81	2026-03-20 03:54:28	26.1759	0.0000	991.8381	1.6667	1875	1
82	2026-03-20 03:54:30	26.1756	0.0000	991.8376	1.6667	1874	1
83	2026-03-20 03:54:32	26.1743	0.0000	991.8387	1.6667	1875	1

83 rows in set (0.00 sec)

mybrano

home about service Dashboard Contact

Live Dashboard

View real-time readings directly from the database. Analyze trends, compare sensors, and check system health all in one place.

Stations: 01 / 01 / 2026, 12:00 AM 12 / 31 / 2026, 11:59 PM

Display Measurements Create Collection

Measurement id	Timestamp	Humidity	Air pressure	Light intensity	Air quality	Station id
1404	2026-03-20 05:22:19	0.0000	990.9324	1.6667	1864	1
1405	2026-03-20 05:22:21	0.0000	990.9381	1.6667	1866	1
1406	2026-03-20 05:22:23	0.0000	990.9391	1.6667	1866	1
1407	2026-03-20 05:22:25	0.0000	990.9286	1.6667	1867	1
1408	2026-03-20 05:22:27	0.0000	990.9362	1.6667	1867	1
1409	2026-03-20 05:22:29	0.0000	990.9240	1.6667	1867	1
1410	2026-03-20 05:22:31	0.0000	990.9338	1.6667	1864	1
1411	2026-03-20 05:22:34	0.0000	990.9310	1.6667	1864	1

Backup

Objective

In this sprint we are going to setup our VM to host our backup server and a automated backup system where a backup of our server generated each certain time using cronjob.

Implementation

Hostname resolution

When you create a MySQL user with a **hostname** like 'backup_user'@'backup', MySQL must be able to resolve that hostname to an IP address (and vice versa) during connection attempts.

What Happens During Connection

When the backup server tries to connect:

1. **Backup server connects** from IP 192.168.4.* to MySQL server
2. **MySQL receives the connection** from IP 192.168.4.*

3. **MySQL performs a reverse DNS lookup** (PTR record):
192.168.4.* → attempts to resolve to 'backup'
4. **MySQL performs a forward DNS lookup** (A record):
'backup' → resolves to 192.168.4.*
5. **MySQL compares:** Does the forward resolution match the original IP?
6. **If YES** → Connection allowed
If NO → Connection rejected with hostname mismatch error

```
ahmab795@pif2026:~$ ping backup
ping: backup: Temporary failure in name resolution
ahmab795@pif2026:~$
```

Edit the hosts files and add ip address of your machines on both machines

Sudo nano /etc/hosts

Ubuntu_Server_PIF Clone (Snapshot 1) [Running] - Oracle VirtualBox	REPIF_backup_VM (Snapshot 1) [Running] - Oracle VirtualBox
<pre>File Machine View Input Devices Help GNU nano 7.2 127.0.0.1 localhost 127.0.1.1 pif2026 # The following lines are desirable for IPv6 capable hosts ::1 ip6-localhost ip6-loopback fe00::0 ip6-localnet ff00::0 ip6-mcastprefix ff02::1 ip6-allnodes ff02::2 ip6-allrouters 192.168.4.70 backup_</pre>	<pre>File Machine View Input Devices Help GNU nano 7.2 127.0.0.1 localhost 127.0.1.1 backup # The following lines are desirable for IPv6 capable hosts ::1 ip6-localhost ip6-loopback fe00::0 ip6-localnet ff00::0 ip6-mcastprefix ff02::1 ip6-allnodes ff02::2 ip6-allrouters 192.168.4.24 pif2026</pre>

Save and try again by pinging by hostname

```
ahmab795@pif2026:~$ nslookup 192.168.4.70
70.4.168.192.in-addr.arpa      name = backup.

ahmab795@backup:~$ nslookup 192.168.4.24
24.4.168.192.in-addr.arpa     name = pif2026.
```

Backup VM installation

- Created a virtual machine in virtualbox
- Ubuntu installed
- Set a hostname
- SSH installed
- Last updates installed

MySQL Backup Configuration

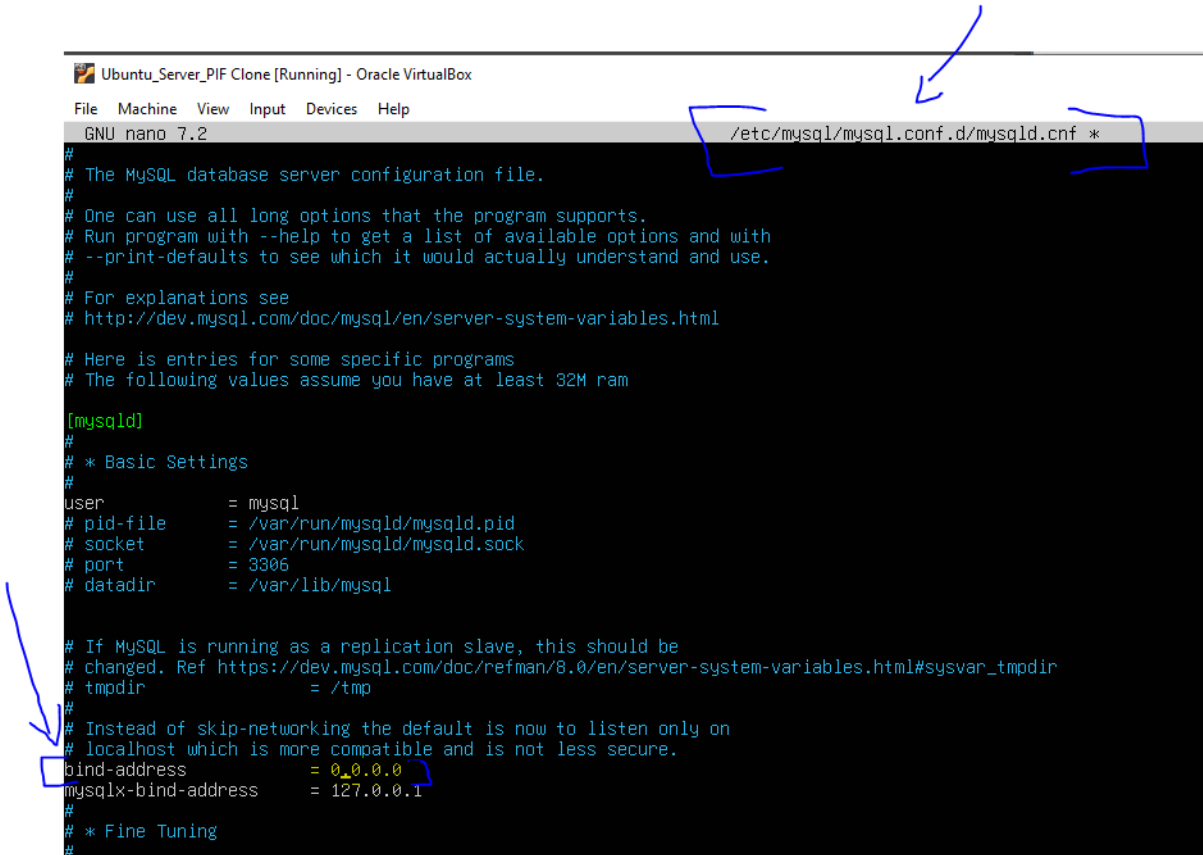
Configure the web/database server and the backup server for database dumps.

Allow External Access:

Open the config file and find the bind-address directive and change it from "127.0.0.1" to "0.0.0.0" to listen all interfaces or specific IP address

Why?

- By default, MySQL only listens on 127.0.0.1 (localhost)
- This means only the same machine can connect to MySQL



```

Ubuntu_Server_PIF Clone [Running] - Oracle VirtualBox
File Machine View Input Devices Help
GNU nano 7.2 /etc/mysql/mysql.conf.d/mysqld.cnf *
#
# The MySQL database server configuration file.
#
# One can use all long options that the program supports.
# Run program with --help to get a list of available options and with
# --print-defaults to see which it would actually understand and use.
#
# For explanations see
# http://dev.mysql.com/doc/mysql/en/server-system-variables.html
#
# Here is entries for some specific programs
# The following values assume you have at least 32M ram
[mysqld]
#
# * Basic Settings
#
user                = mysql
# pid-file           = /var/run/mysqld/mysqld.pid
# socket             = /var/run/mysqld/mysqld.sock
# port               = 3306
# datadir            = /var/lib/mysql

# If MySQL is running as a replication slave, this should be
# changed. Ref https://dev.mysql.com/doc/refman/8.0/en/server-system-variables.html#sysvar_tmpdir
# tmpdir             = /tmp
#
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
# bind-address        = 0.0.0.0
# mysqlx-bind-address = 127.0.0.1
#
# * Fine Tuning
#

```

This command ensure us the configuration we made is correct and mysql listening on network

```

ahmab795@pif2026:~$ sudo netstat -tlnp | grep 3306
tcp        0      0 0.0.0.0:3306          0.0.0.0:*           LISTEN      1065/mysqld
tcp        0      0 127.0.0.1:3306      0.0.0.0:*           LISTEN      1065/mysqld
ahmab795@pif2026:~$

```

then restart MySQL

```

ahmab795@pif2026:~$ sudo systemctl restart mysql
ahmab795@pif2026:~$

```

Firewall

Double check the firewall to allow traffic coming from 3306

```

ahmab795@backup:~$ sudo ufw allow 3306/tcp
[sudo] password for ahmab795:
Sorry, try again.
[sudo] password for ahmab795:
Rules updated
Rules updated (v6)

```


Restricted MySQL User(Database server side)

A MySQL user with minimum privileges to backup the project database.

```
mysql> create user 'backup_user'@'backup' identified by 'AhmAb795';
Query OK, 0 rows affected (0.04 sec)

mysql> grant lock tables,select,show view on PIF_2026.* TO 'backup_user'@'backup';
Query OK, 0 rows affected (0.07 sec)

mysql> flush privileges;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

The image above represents creation of a user with username 'backup_user' and connection allowed from 'backup' host.

Install MySQL Client(backup server side)

Run the following command to install latest version of MySQL Client, needed for *mysqldump*

Sudo apt update
Sudo apt install mysql-client -y

Backup Directory Creation

Create a directory for backups and assign appropriate permissions.

```
ahmab795@backup:~$ sudo mkdir -p /var/backups/PIF_2026
ahmab795@backup:~$ sudo chmod 750 /var/backups/PIF_2026
ahmab795@backup:~$
```

Secure MySQL Password

Use `mysql_config_editor` to store the credentials for the backup_user securely. This avoids hardcoding passwords in scripts.

```
ahmab795@backup:~$ mysql_config_editor set --login-path=local_backup --host=pif2026 --user=backup_user --password
Enter password:
```

The credentials are now stored in an encrypted file (`~/.mylogin.cnf`).

Test basic connectivity

```
ahmab795@backup:~$ mysql -h pif2026 -u backup_user -p -e "select user(), current_user();"
Enter password:
+-----+-----+
| user() | current_user() |
+-----+-----+
| backup_user@backup | backup_user@backup |
+-----+-----+
```

Backup Shell Script

Create a script in with the following content:

```
ahmab795@backup: ~
GNU nano 7.2 /usr/local/bin/backup_PIF_2026_db.sh
#!/bin/bash
# Database backup script

# Configuration
BACKUP_DIR="/var/backups/PIF_2026"
DATABASE_NAME="PIF_2026"
LOG_FILE="/var/log/PIF_2026_backup_db.log"
DATE=$(date +%Y%m%d_%H%M%S)
BACKUP_FILE="$BACKUP_DIR/${DATABASE_NAME}_${DATE}.sql"

# Log start
echo "$(date): Backup started." >> $LOG_FILE

# Perform the backup using mysql_config_editor login path
if mysqldump --login-path=local_backup --single-transaction $DATABASE_NAME > $BACKUP_FILE 2>> $LOG_FILE; then
    echo "$(date): Backup successful. File: $BACKUP_FILE" >> $LOG_FILE
    # Optional: Compress the backup
    gzip $BACKUP_FILE
    echo "$(date): Backup compressed." >> $LOG_FILE
else
    echo "$(date): Backup FAILED." >> $LOG_FILE
    exit 1
fi

# Delete backups older than 7 days (Could Requirement)
find $BACKUP_DIR -name "${DATABASE_NAME}*.sql.gz" -type f -mtime +7 -delete
echo "$(date): Old backups cleaned up." >> $LOG_FILE

echo "$(date): Backup script finished." >> $LOG_FILE
exit 0
```

Save and make it executable.

```
ahmab795@backup:~$ sudo chmod +x /usr/local/bin/backup_PIF_2026_db.sh
ahmab795@backup:~$ sudo ls /usr/local/bin/
backup_PIF_2026_db.sh
ahmab795@backup:~$
```

Test the script

Run the script to pull a backup from webserver and check if saves correctly.

```
ahmab795@backup:~$ /usr/local/bin/backup_PIF_2026_db.sh
ahmab795@backup:~$ ls /var/backups/PIF_2026/
PIF_2026_20260511_083840.sql.gz
```

Schedule the backup

Open your crontab and add the following command in the end.

```
Sudo crontab -e
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 8 * * * /usr/local/bin/backup_PIF_2026_db.sh_
```

Save and exit

FTP configuration

A FTP server and a user need to be created in order to save and make backups accessible.

1) FTP server overview

Item	Detail
FTP server software	Vsftpd
Server hostname	Backup
Server current IP address	192.168.188.5
FTP port	21(standard)
Backup directory	/var/backups/

2) FTP user configuration

Item	Detail
FTP username	Backup_ftp_user
Authentication	AhmAb795
Home directory	/var/backups/
Sell	nologin

3) Vsftpd configuration file

Location: /etc/vsftpd.conf

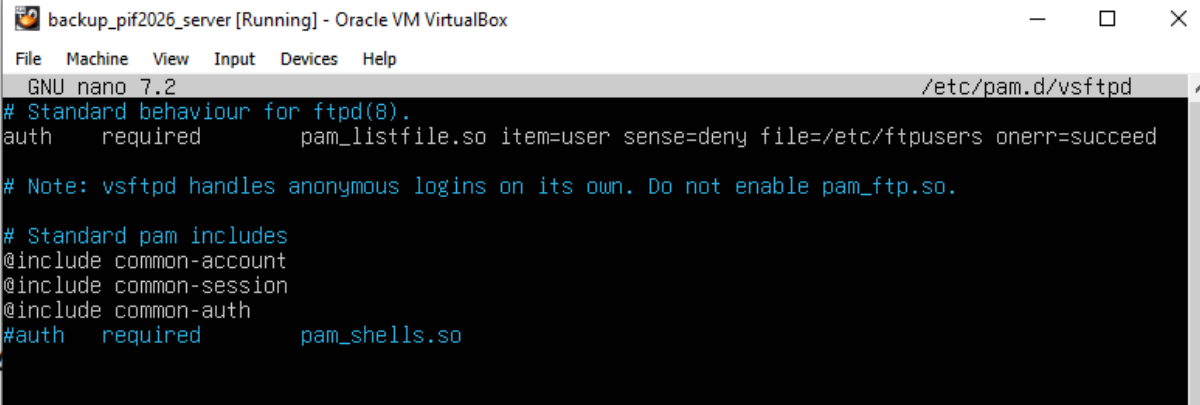
Key configuration settings:

- Local_enable=YES
- Write_enable=YES
- Chroot_local_user=YES
- Allow_writeable_chroot=YES
- Anonymous_enable=NO
- Listen_ipv6=NO
- Listen=YES

4) PAM configuration modification

Location: /etc/pam.d/vsftpd

Comment the following out



```

backup_pif2026_server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
GNU nano 7.2 /etc/pam.d/vsftpd
# Standard behaviour for ftpd(8).
auth required pam_listfile.so item=user sense=deny file=/etc/ftpusers onerr=succeed

# Note: vsftpd handles anonymous logins on its own. Do not enable pam_ftp.so.

# Standard pam includes
@include common-account
@include common-session
@include common-auth
#auth required pam_shells.so

```

5) Directory permissions

```

ahmab795@backup:~$ sudo chown backup_ftp_user:backup_ftp_user /var/backups
ahmab795@backup:~$ sudo chmod 755 /var/backups

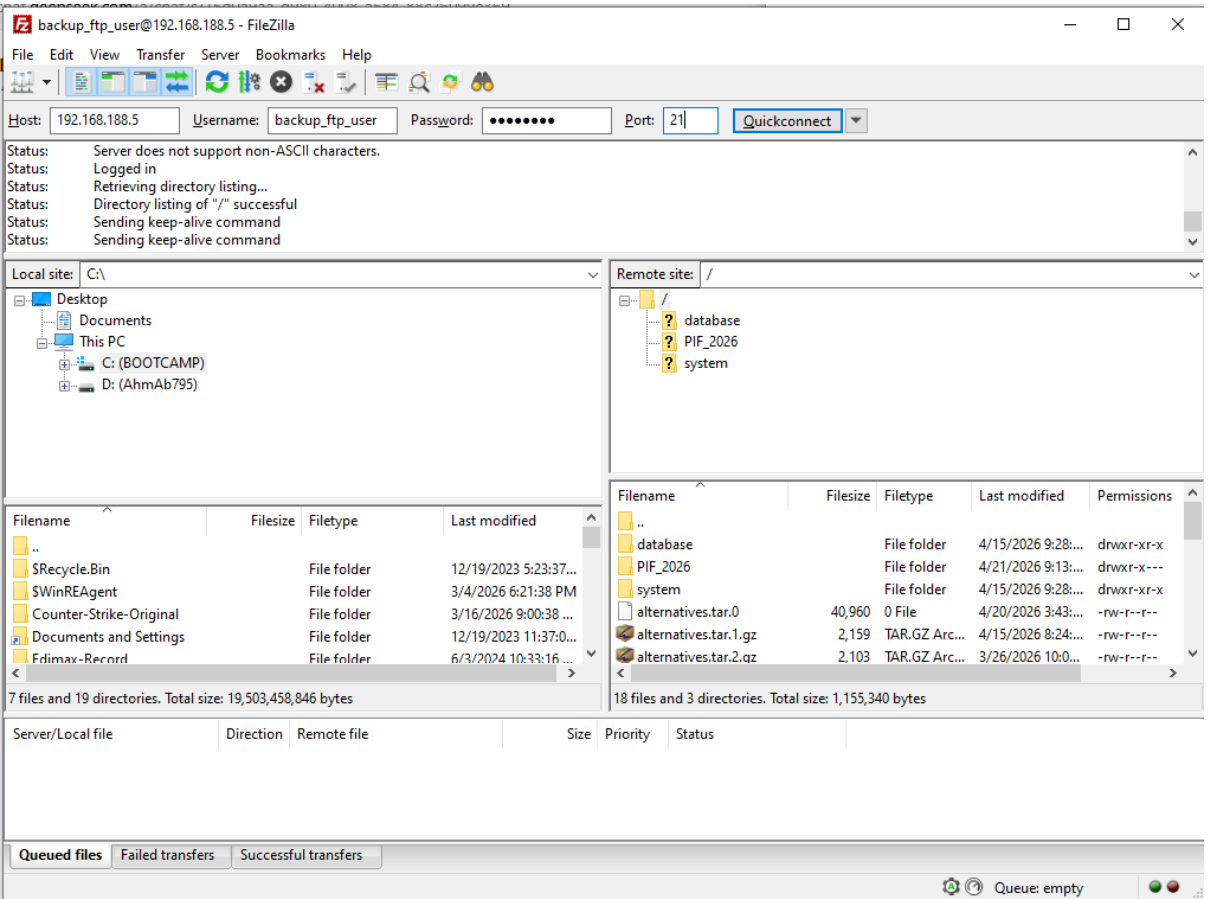
```

6) Test FTP access locally and remotely

```

ahmab795@backup:~$ ftp localhost
Connected to localhost.
220 Welcome from Abolo to FTP service.
Name (localhost:ahmab795): backup_ftp_user
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.

```



FileZilla interface showing a successful FTP connection to 192.168.188.5. The local site is C:\ and the remote site is /. The interface shows file listings for both sites and a status bar at the bottom indicating 'Queue: empty'.

Filename	Filesize	Filetype	Last modified	Permissions
..				
database		File folder	4/15/2026 9:28:...	drwxr-xr-x
PIF_2026		File folder	4/21/2026 9:13:...	drwxr-xr-x
system		File folder	4/15/2026 9:28:...	drwxr-xr-x
alternatives.tar.0	40,960	0 File	4/20/2026 3:43:...	-rw-r--r--
alternatives.tar.1.gz	2,159	TAR.GZ Arc...	4/15/2026 8:24:...	-rw-r--r--
alternatives.tar.2.qz	2,103	TAR.GZ Arc...	3/26/2026 10:0:...	-rw-r--r--

7) FTP connection in terminal(WebS -> Backups)

Basic FTP connection

```

ftp 'your-server-ip/hostname'

```

```

ahmab795@pif2026:~$ ftp 192.168.6.179
Connected to 192.168.6.179.
220 Welcome from Abolo to FTP service.
Name (192.168.6.179:ahmab795): backup_ftp_user
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> cd PIF_2026
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||52965|)
150 Here comes the directory listing.
-rwxr-xr-x 1 0 1001 0 Apr 30 09:25 PIF_2026_DB_backup_20260430_092548.sql
-rwxr-xr-x 1 0 1001 0 May 04 09:08 PIF_2026_DB_backup_20260504_090845.sql
-rw-r--r-- 1 0 0 0 May 07 09:06 PIF_2026_DB_backup_20260507_090611.sql
-rw-r--r-- 1 0 0 0 May 07 09:22 PIF_2026_DB_backup_20260507_092259.sql
-rw-r--r-- 1 0 0 0 May 07 09:23 PIF_2026_DB_backup_20260507_092338.sql
226 Directory send OK.
ftp> user 'get' to download and 'put' to upload file

```

Use “get <File Name> <path to save>” to download desire file.

```
ftp> get PIF_2026_20260511_083840.sql.gz
local: PIF_2026_20260511_083840.sql.gz remote: PIF_2026_20260511_083840.sql.gz
229 Entering Extended Passive Mode (|||37789|)
150 Opening BINARY mode data connection for PIF_2026_20260511_083840.sql.gz (287
38 bytes).
100% |*****| 2878      51.78 MiB/s    00:00 ETA
226 Transfer complete.
2878 bytes received in 00:00 (1.40 MiB/s)
ftp> exit
221 Goodbye.
ahmab795@pif2026:~$ ls
PIF_2026_20260511_083840.sql.gz  test_backup.sql
ahmab795@pif2026:~$ gunzip PIF_2026_20260511_083840.sql.gz
ahmab795@pif2026:~$ ls
PIF_2026_20260511_083840.sql  test_backup.sql
ahmab795@pif2026:~$
```

Installation in Different Network

The system may need to be installed and run on different networks. This section addresses the necessary adjustments to ensure the system functions properly regardless of the network environment.

Key Considerations for Cross-Network Compatibility

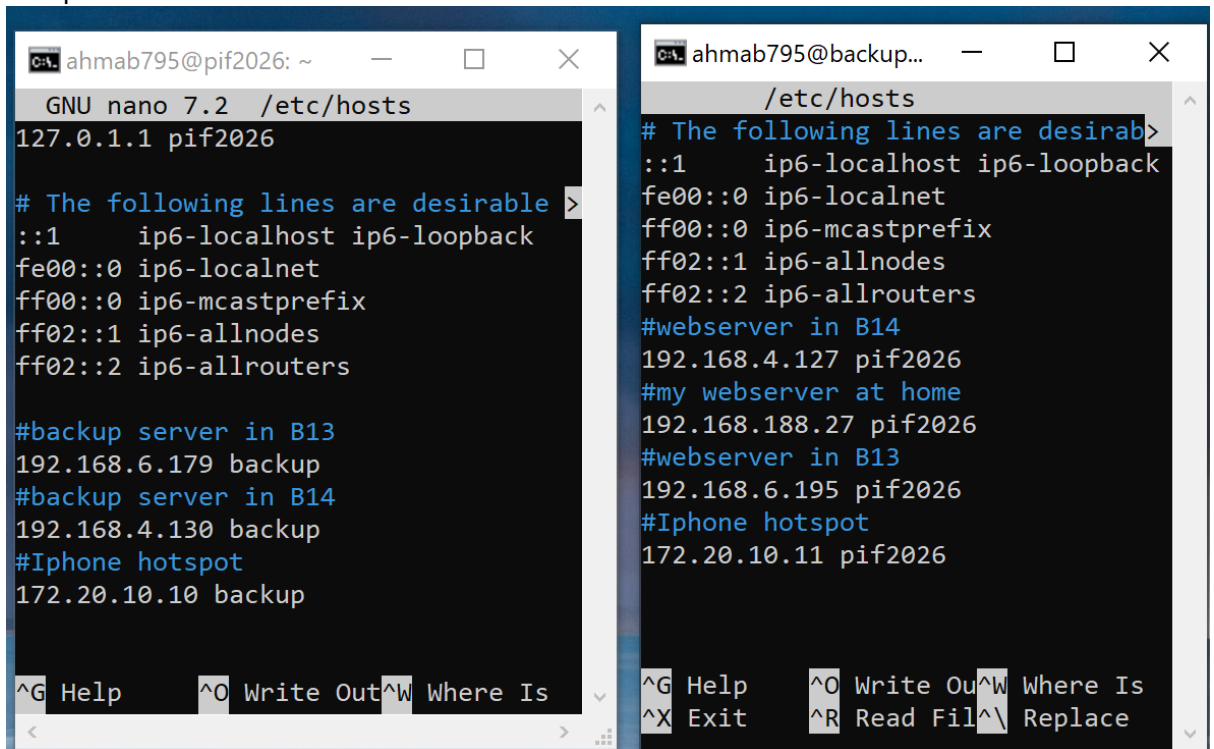
When deploying the system on a different network, the database connection settings must be updated:

Parameter	Typical Value	Adjustment Required	Image reference
Raspberry Pi	Connection to network	Physically connect a keyboard and mouse, then connect to a network	
Host resolution	IP addresses	Provide both servers' IP addresses	Figure13)host resolution
Username and password	Username: Webserver_user Pass: AhmAb795??	The database username and password for a special user must be defined in db_config.php	Figure14)DB access
Destination Sent Measurement	IP address	In the station, /opt/station/config.toml must include the web server's address and path (to DataHandler.php)	Figure15))Data Transferring

Example Configuration image approach:

1. Host resolution

Converting a hostname (e.g., web-server) into an IP address (e.g., 192.168.1.10) so computers can find each other on the network.



```

GNU nano 7.2 /etc/hosts
127.0.1.1 pif2026

# The following lines are desirable
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

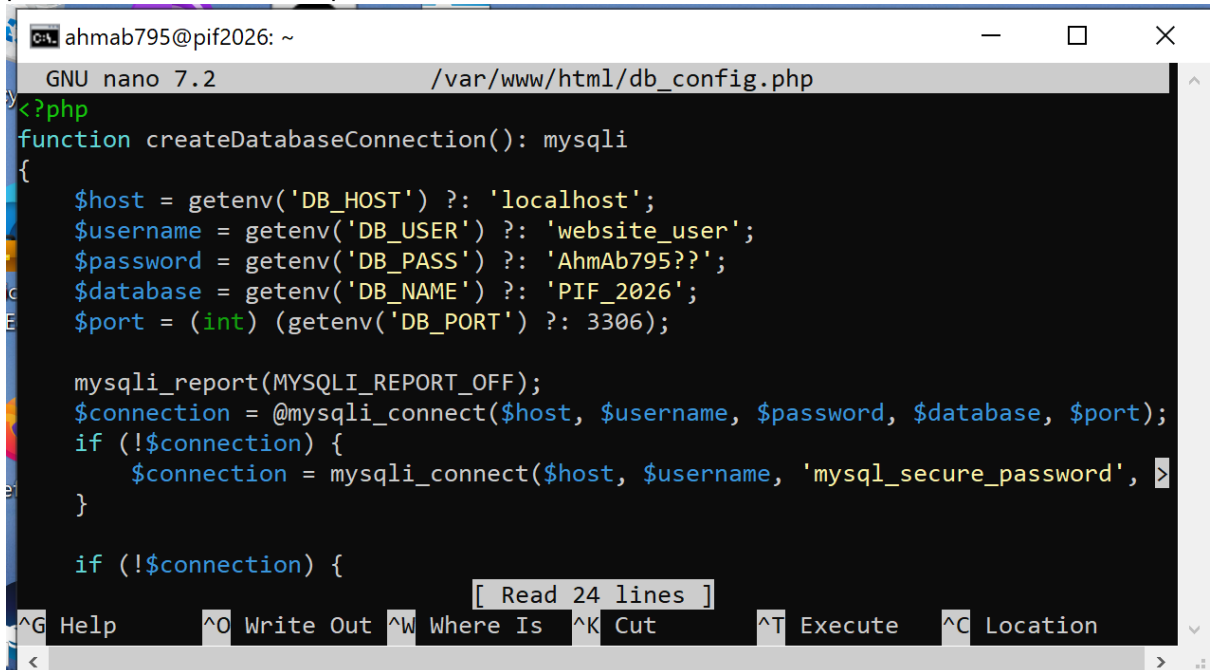
#backup server in B13
192.168.6.179 backup
#backup server in B14
192.168.4.130 backup
#Iphone hotspot
172.20.10.10 backup

GNU nano 7.2 /etc/hosts
# The following lines are desirable
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
#webserver in B14
192.168.4.127 pif2026
#my webserver at home
192.168.188.27 pif2026
#webserver in B13
192.168.6.195 pif2026
#Iphone hotspot
172.20.10.11 pif2026
  
```

Figure 13) host resolution

2. Database Connection

Web user authentication credentials must be stored in **db_config.php** for the application to connect to the database. Since we don't use environment variable we provide the credentials in plain text.



```

GNU nano 7.2 /var/www/html/db_config.php
<?php
function createDatabaseConnection(): mysqli
{
    $host = getenv('DB_HOST') ?: 'localhost';
    $username = getenv('DB_USER') ?: 'website_user';
    $password = getenv('DB_PASS') ?: 'AhmAb795??';
    $database = getenv('DB_NAME') ?: 'PIF_2026';
    $port = (int) (getenv('DB_PORT') ?: 3306);

    mysqli_report(MYSQLI_REPORT_OFF);
    $connection = @mysqli_connect($host, $username, $password, $database, $port);
    if (!$connection) {
        $connection = mysqli_connect($host, $username, 'mysql_secure_password',
    }

    if (!$connection) {
  
```

Figure 14)DB access

3. Transferring Data(Raspberry Pi)

Provide IP address and path of your Webserver in “config.toml” to transfer captured data to right destination.

```
ahmab795@raspberrypi: ~  
GNU nano 8.4 /opt/station/config.toml  
station_serial = "WST-202601-001" # Serial number of the station  
  
interval = 2 # Time between measurements  
  
toggle_intensity = 25 # The light level at which the LED should be turned  
  
[i2c] # I2C-Addresses of the sensors  
bme280 = 0x76  
bh1750 = 0x23  
mq135 = 0x48  
  
[destination] # Information about the server  
ip = "192.168.6.195" # IP-Address or Hostname of the Server  
path = "/DataHandler.php" # Path to the script that accepts the measurements
```

Figure 15)Data Transferring